



Deep learning based visually rich document content understanding: a survey

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Abstract

Visually Rich Documents (VRDs) play a vital role in domains such as academia, finance, healthcare, and marketing, as they convey information through a combination of text, layout, and visual elements. Traditional approaches to extracting information from VRDs rely heavily on expert knowledge and manual annotation, making them labor-intensive and inefficient. Recent advances in deep learning have transformed this landscape by enabling multimodal models that integrate vision, language, and layout features through pretraining, significantly improving information extraction performance. This survey presents a comprehensive overview of deep learning-based frameworks for VRD Content Understanding. We categorize existing methods based on their modeling strategies and downstream tasks, and provide a comparative analysis of key components, including feature representation, fusion techniques, model architectures, and pretraining objectives. Additionally, we highlight the strengths and limitations of each approach and discuss their suitability for different applications. The paper concludes with a discussion of current challenges and emerging trends, offering guidance for future research and practical deployment in real-world scenarios.

Keywords Visually rich document · Deep learning · Multimodal learning

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1 Introduction

1.1 Backgrounds

Visually Rich Documents (VRDs) are widely used across various domains, including finance, healthcare, law, education, and research. Both structured and unstructured visual-textual information must be interpreted in combination to derive meaningful insights. Unlike plain-text documents, VRDs contain heterogeneous components such as paragraphs, tables, charts, diagrams, and images, which are intricately arranged in formats like PDFs (Ding et al. 2024, 2023), Word documents (Li et al. 2020), or scanned images (Ding et al. 2025). These visual and textual elements, referred to as document **semantic entities**, play a critical role in summarizing, illustrating, or substantiating key content, often through complex layouts and visual cues.

Understanding VRDs presents significant challenges due to their semi-structured or unstructured nature, which limits the effectiveness of traditional rule-based or linear text processing methods. This complexity has given rise to Visually Rich Document Content Understanding (VRD-CU), a specialized field focused on automatically extracting, interpreting, and reasoning over such documents to enable machine-readability and scalable knowledge extraction.

Historically, early VRD-CU approaches were heuristic-driven, relying on layout analysis and domain-specific templates (Watanabe et al. 1995; Seki et al. 2007), later expanding to include statistical models (Rusinol et al. 2013; Oliveira and Viana 2017). However, these methods required manually crafted rules, lacked flexibility, and often failed to generalize beyond fixed document structures or domains. The emergence of deep learning marked a turning point in VRD-CU. Pioneering models such as Chargrid (Katti et al. 2018) and BERTgrid (Denk and Reisswig 2019) applied convolutional and recurrent neural networks to treat document layouts as two-dimensional text grids. This led to the development of feature-based models such as PICK (Yu et al. 2021), TRIE (Zhang et al. 2020), and Ephoie (Wang et al. 2021), which integrated layout features with textual signals for improved extraction.

To acquire comprehensive VRD representations and improve generalization across VRD tasks, BERT-style pretrained frameworks have been proposed to achieve comprehensive document representation in VRD. They align multimodal information through various pre-training tasks. However, their focus on fine-grained word sequences limits the ability to capture higher-level semantic relationships. To address these limitations, coarse-grained models are introduced (Gu et al. 2021; Li et al. 2021a) that focus on entity-level or semantic block-level understanding, offering stronger abstraction capabilities while sacrificing some fine-grained detail. Striking a balance between detail and abstraction, joint-grained frameworks such as StructExt (Li et al. 2021b), StructExtV2 (Yu et al. 2022), and WuKong (Bai et al. 2022) have emerged, combining both granular and holistic representations. More recently, Large Language Model (LLM)-based architectures—including LayoutLLM (Luo et al. 2024) and HRVDA (Liu et al. 2024)—have been proposed to support cross-modal reasoning and complex semantic understanding through instruction tuning and prompt-driven paradigms.

In this survey, we provide a comprehensive review of deep learning-based VRD-CU methodologies, tracing their evolution from early neural encoders to current multimodal

transformers and LLM-based frameworks. We categorize existing approaches based on task formulation, model architecture, feature integration strategies, and pretraining objectives, comparing their effectiveness, generalisability, and scalability across benchmark datasets. We further highlight emerging trends, challenges, and future directions, aiming to support both researchers and practitioners in navigating this dynamic and impactful research area.

1.2 Scope

This survey aims to provide a comprehensive and critical review of recent advances in **deep learning-based Visually Rich Document Content Understanding (VRD-CU)**. We focus on approaches that leverage neural architectures to understand and reason over the content of complex VRDs. The scope of this survey is defined by the following criteria:

- *Focus on Content-Level Understanding Tasks* We include works that address *document content understanding* tasks, specifically Key Information Extraction (KIE), Visually Rich Document Question Answering (VRD-QA), Entity Linking (EL). We **exclude** studies that are limited to *document structure analysis* (e.g., layout segmentation or reading order detection) without directly contributing to semantic content understanding.
- *Deep Learning-based Models* We primarily review papers that propose novel *deep learning methods* tailored for VRD-CU tasks. While Sect. 2.2.1 briefly summarises early heuristic and traditional machine learning approaches for historical context, the core focus remains on modern neural frameworks.
- *Dataset Coverage* We compile and analyse publicly available *benchmark datasets* for VRD-CU tasks, particularly those introduced or widely adopted in *top-tier conferences and journals since 2019*. These datasets support content-level reasoning in multimodal documents.
- *Emphasis on Multimodal and Multi-page Understanding* In contrast to previous surveys that focus narrowly on subtasks or single-page settings, this paper highlights models capable of *multimodal reasoning* and handling *entire, multi-page documents holistically*, better reflecting real-world document comprehension needs.

This survey is designed to support researchers and practitioners in understanding the current landscape of deep learning models for VRD-CU, identifying emerging trends, and navigating the key resources, including models and datasets, which define this growing field. It should be noted that this article only investigates models and datasets designed for the entire document or the document page rather than the specific document components such as *table*, *chart*. Therefore, models and datasets proposed for table detection, table structure recognition, and chart or plot question answering will not be summarized in the main body of this survey.

1.3 Related surveys

Several surveys offer a comprehensive overview of general document understanding tasks. These surveys primarily focus on document layout analysis (Subramani et al. 2020), table extraction (Liu et al. 2023), named entity recognition (Ehrmann et al. 2023), and document

image analysis (Lombardi and Marinai 2020) across diverse document types, including invoices (Saout et al. 2024) and historical document (Ehrmann et al. 2023; Lombardi and Marinai 2020). In particular, computer vision-based research focuses on scanned document analysis and structure understanding. While these studies have advanced document image analysis, they often focus on fragmented subtasks and fall short of providing a holistic understanding of an entire document on multiple pages.

Recent advancements in deep learning have fueled the emergence of VRD-CU tasks, which demand complex document content understanding capabilities. These tasks include key information extraction, question answering, and document entity linking. However, existing surveys have not adequately addressed the unique challenges and opportunities presented by VRD-CU, with a specific emphasis on deep learning-based multimodal approaches (Cui et al. 2021). To bridge this gap, this survey aims to provide a comprehensive overview of VRD-CU frameworks and datasets, including multimodal feature extractions and fusions in both mono and multi-task VRD models.

1.4 Contributions

The main contributions of this paper can be summarized as follows:

- The paper provides a detailed review and systematic categorization of VRD-CU frameworks and benchmark datasets, organized based on adopted strategies and downstream tasks.
- It critically examines and compares different techniques used in VRD-CU models, focusing on feature representation and fusion, model architecture, and pretraining methods, highlighting their strengths, limitations, and appropriate scenarios.
- The paper identifies emerging trends and challenges in visually rich document content understanding, offering insights into future research directions and practical applications.

1.5 Survey structure

Section 1 introduces the background of VRD-CU, outlining the aim, scope, and key contributions of this survey. Section 2 provides essential background knowledge, including definitions of the VRD-CU tasks, the evolution of document understanding techniques and terminology definition. Sections 3 and 4 review notable mono-task and multi-task frameworks, respectively. Section 5 presents benchmark datasets across three VRD-CU subtasks with evaluation metrics description. Section 6 offers a critical analysis of the reviewed models, highlighting their strengths and limitations through quantitative evaluation. Section 7 identifies emerging trends and key challenges, with a focus on practical applications. Finally, Sect. 8 summarizes the survey findings.

2 Background

2.1 Task definition

Based on their purpose and application scenarios, visually rich document content understanding (VRD-CU) tasks can be categorized into three main types: **Key Information Extraction**, **Question Answering**, and **Entity Linking**, as illustrated in Fig. 1.

- *Key Information Extraction (KIE)* refers to identifying and extracting the relevant information based on the given text queries. Distinct pre-defined queries can be defined based on the domain of targeting documents and practical demands. For example, the crucial information of scanned receipts contains “Store Name”, “Address”, “Item” and “Price”, while for the financial reports, “Company Name”, “Share Holder Name”, “Number of Interests” may be the critical information need to be extracted.
- *Entity Linking (EL)* refers to identifying the semantic relations between document semantic entities to construct the logical structure of the input document image.
- *Question Answering (VRD-QA)* the task of answering natural language questions about visual-rich documents (VRDs), first formalized by the DocVQA (Mathew et al. 2021) dataset. Depending on the answer type, it can be categorized into extractive QA, where answers are directly extracted from the document, and generative QA, which requires producing answers by comprehensively understanding both the questions and the associated VRDs.

2.2 Development of document understanding

2.2.1 Traditional approaches

Rule-based methods, as highlighted in several studies (Watanabe et al. 1995; O’Gorman 1993; Seki et al. 2007; Rusinol et al. 2013), have demonstrated high precision in domain-specific applications. However, these methods have several drawbacks: they are manually intensive, costly, and require expert intervention for tailored customization. Additionally, they are inflexible, often necessitating frequent manual updates, even for minor modifications. In response to these limitations, machine learning-based approaches have been proposed for document understanding. For instance, SVM-based methods have been utilized for layout understanding (Oliveira and Viana 2017), and TF-IDF techniques combined with hand-crafted features have been applied for extracting information from invoices. Further-

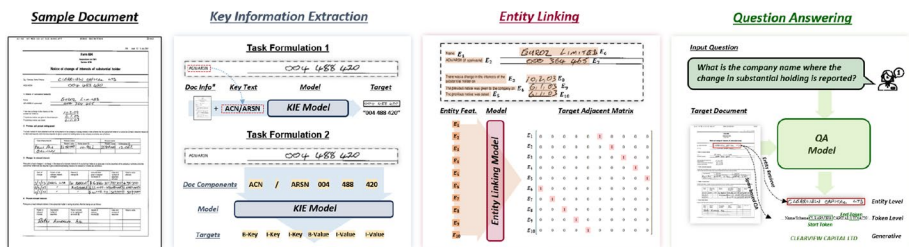


Fig. 1 Visually rich document content understanding task clarifications

more, rule-based and statistical models have been used for entity extraction. Despite these advancements, machine learning methods rely heavily on human intervention and domain-specific expertise. They are time-consuming and often deliver suboptimal performance. Moreover, the majority of these methods typically depend on single-modality data, restricting them to either layout, text, or visual inputs.

2.2.2 Single modality-based approaches

With the advancement of deep learning, deeper model architectures such as CNNs (Katti et al. 2018; Yang et al. 2017) have emerged, and pretrained language (Devlin 2018; Liu et al. 2019) and vision models (Ren et al. 2015; He et al. 2017) are now commonly used as robust baselines for understanding VRD content and structure. Due to the multimodal nature of VRDs, which involves the integration of text, vision, and layout, researchers are increasingly focusing on leveraging this combined information to achieve significant improvements in various downstream VRD-CU tasks.

2.2.3 Cross modality based approaches

Considering the multimodal nature of VRDs, many frameworks propose various methods to encode multimodal information, including text, vision, and layout, and fuse them effectively. Text and vision information is normally encoded by pretrained backbones such as BERT (Devlin 2018) or RoBERTa (Liu et al. 2019) for textual features, Faster-RCNN (Ren et al. 2015) or Mask-RCNN (He et al. 2017) for visual features. For layout information, different encoding methods are introduced, including linear projection (Wang et al. 2020), 2D positional encoding (Xu et al. 2020a), and attention bias to allow the proposed models to be layout-sensitive. Different feature fusion methods are introduced including summing up (Yu et al. 2021), concatenation (Lee et al. 2022), attention-based contextual learning (Majumder et al. 2020), and prompting (He et al. 2023). However, most of those frameworks leverage implicit knowledge from pretrained backbones with a task-orientated shadow adapter for specific VRD-CU downstream tasks such as KIE (Lee et al. 2023; Wang and Shang 2022; Chen et al. 2023; Cao et al. 2023) or EL (Zhang et al. 2021; Hu et al. 2023; Carbonell et al. 2021). Those frameworks tend to achieve delicate performance on specific tasks or document formats instead of acquiring a generalized model to represent documents comprehensively.

2.2.4 Multimodal pretraining approaches

Inspired by the success of BERT-style models (Devlin 2018; Liu et al. 2019) in acquiring knowledge through self-supervised learning, pretrained document understanding models have emerged to harness self-supervised or supervised pretraining tasks from extensive document collections. LayoutLM (Xu et al. 2020a), the first encoder-only VRD-CU model, utilizes self-supervised tasks, such as masked vision-language modeling, with text and layout information. Subsequent models have expanded on this by integrating layout information (Wang et al. 2022; Tu et al. 2023; Li et al. 2021b) and visual cues (Xu et al. 2020b; Huang et al. 2022) through multimodal transformers. While encoder-only models have shown significant improvements on various benchmark datasets (Jaume et al. 2019; Park et al. 2019;

Huang et al. 2019; Mathew et al. 2021; Harley et al. 2015), they often require detailed annotations and are limited by fixed input lengths. To address these limitations, encoder–decoder frameworks (Tang et al. 2022; Kim et al. 2022; Davis et al. 2022) and prompt-based methods for LLMs/MLLMs (Luo et al. 2024; Liu et al. 2024) have been developed, enhancing layout awareness and performance in VRD-CU tasks. However, a significant gap remains in effectively applying these models in real-world scenarios with zero shot.

2.3 Terminology definition

After reviewing the evolution of VRD-CU, we introduce standardized terminology to categorize existing frameworks along three key dimensions: **Downstream Task Focus**, **Model Architecture**, **Textual Granularity** and **Visual Information**.

- *Downstream Task Focus* Frameworks are categorized into *mono-task* and *multi-task* types. Mono-task frameworks (see Sect. 3) are tailored for a single VRD-CU task (e.g., KIE or VQA) with task-specific designs, while multi-task frameworks (see Sect. 4) are designed to handle multiple tasks simultaneously, typically leveraging pretraining or shared components to improve generalization ability.
- *Model Architecture* Frameworks can be grouped into *encoder-only*, *encoder–decoder*, and *decoder-only* categories. Encoder-only models focus on representation learning; encoder–decoder models generate task-specific outputs from document inputs; decoder-only models use auto-regressive decoding for end-to-end generation.
- *Textual Granularity* Depending on how document structure is represented, frameworks process inputs at different levels: **fine-grained** (word-level), **coarse-grained** (e.g., entity-level inputs such as paragraphs, tables, or section headers), or **joint-grained**, which combines both. Fine-grained inputs capture detailed spatial and lexical information through individual word tokens, while coarse-grained representations emphasize higher-level semantic structures and contextual dependencies.
- *Visual Information* For frameworks that incorporate visual information, we categorize them based on the visual feature extraction methods: **RoI-based** and **patch-based**. RoI-based frameworks often use object detectors or document parsers to identify Regions of Interest (RoIs) corresponding to semantic entities, followed by a CNN (e.g., ResNet) to extract visual features via RoI-Align—commonly referred to as *RoI-based* frameworks. In contrast, Transformer-based frameworks (e.g., ViT Dosovitskiy et al. (2020), Swin-Transformer Liu et al. (2021)) typically divide the document image into uniform patches and encode them sequentially, forming the basis of *patch-based* frameworks.

These terms provide a unified lens to understand and compare the design choices across VRD-CU frameworks in Sects. 3, 4 and 6.

3 Mono-task document understanding frameworks

VRD-CU comprises several distinct downstream tasks, each designed to meet specific application needs and user requirements. This section introduces methods tailored to individual tasks. As illustrated in Fig. 2, we focus on three core VRD-CU tasks—**Key Information**

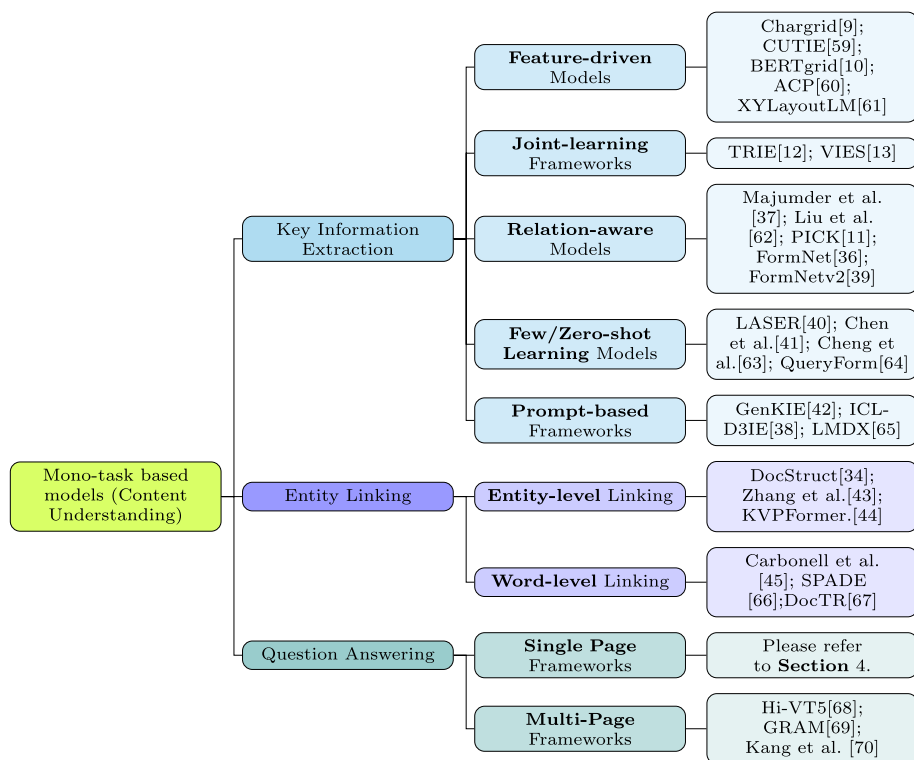


Fig. 2 Mono-task visually rich document understanding models

Extraction (KIE), Entity Linking (EL), and Question Answering (QA)—and summarize corresponding models along with key trends and insights in their development.

3.1 Key information extraction

Key Information Extraction (KIE), a typical natural language processing task, refers to the task of identifying and extracting crucial pieces of information from textual data. Unlike typical name entity recognition methods for addressing plain text, VRDs contain visually rich entities like *tables* and *charts*, as well as spatial and logical layout arrangements to enhance the challenge of extracting crucial information. Although plain-text pretrained language models, such as BERT (Devlin 2018), RoBERTa (Liu et al. 2019), and ALBERT (Lan et al. 2019), are widely used as solid baselines on many benchmark datasets, more recent works introduce layout-aware pretrained models such as LayoutLM families (Xu et al. 2020a, b; Huang et al. 2022), LiLT (Wang et al. 2022), Bros (Hong et al. 2021) to enhance the document representation by leveraging visual and layout information and achieving SoTA performance on several downstream tasks (see Sect. 4). This section will mainly focus on models specifically proposed for the document KIE models or only evaluated on KIE benchmark datasets including FUNSD (Jaume et al. 2019), CORD (Park et al. 2019), SROIE (Huang et al. 2019), XFUND (Xu et al. 2021), etc. Based on innovative aspects, we categorize KIE frameworks into five types: *Feature-driven models* use multimodal cues for

rich feature representation. *Joint Learning frameworks* integrate auxiliary tasks to enhance document representation. *Relation-aware models* leverage spatial or logical relations via graphs or masked attention mechanisms. *Few/Zero-shot learning frameworks* explore methods for extracting key information with minimal labeled data, often using transfer learning. *Prompt-based frameworks* use structured prompts to guide specific information extraction from pretrained models or LLM/MLLMs. These categories represent diverse approaches to improving KIE by leveraging specific model designs and learning strategies tailored to document understanding challenges.

3.1.1 Feature-driven models

In the initial phase, certain Recurrent Neural Network (RNN)-based models (Lample et al. 2016) were introduced, primarily focusing on key information extraction tasks from plain text. However, these approaches overlook the importance of visual cues and layout information. Hence, several multimodal frameworks with feature-driven designs have been proposed to generate more representative document representations.

Chargrid (Katti et al. 2018) first mentioned the significance of 2D structure for document KIE and designed a character-box-based chargrid to convert the textual and 2D layout structure into colored **visual cues** feed into CNN. Instead of using fine-grained character information, **CUTIE** (Zhao et al. 2019) and **BERTgrid** (Denk and Reisswig 2019) utilize various word embedding methods with bounding box coordinates to persevere the layout structure. **ACP** (Palm et al. 2019) leverages attention mechanism and multi-aspect features, including visual, semantic (both character and word) and spatial, into dilated CNN for capturing both short and long-term dependencies of each word piece.

Joint learning KIE frameworks are proposed to leverage multi-level features to mitigate the information gap between various focused tasks. **TRIE** (Zhang et al. 2020) firstly offers an end-to-end framework for simultaneously conducting Optical Character Recognition (OCR) and KIE. OCR module will generate multi-aspect features, including positional, visual and textual aspects. Adaptively trainable weighting mechanisms are adopted to generate the fused embedding, followed by a Bi-LSTM-based Entity Extraction Module to conduct the final prediction. **VIES** (Wang et al. 2021) is introduced to use the multi-level cues of vision, position and text to generate more comprehensive representations. Both token and segment (entity) level positional and visual features are acquired from text detection modules, and dual-level textual features are gathered by text recognition brunch, fused by a self-attention-based fusion module for sequence labeling.

3.1.2 Relation-aware models

Beyond leveraging multimodal signals, modeling the spatial and logical relationships between document components can yield more robust and context-aware representations. Relation-aware models explicitly encode these inter-element dependencies—often using graph structures—to support complex information extraction tasks. Majumder et al. (2020) propose a two-stage approach: field-specific detectors first generate candidates, which are then scored by a neural model that encodes both textual and spatial features. This enables the model to adapt across varying document layouts by focusing on field relevance.

Graph-based frameworks have gained traction for encoding spatial/logical relationships. Liu et al. (2019) pioneer this by constructing a fully connected graph where each node is a textual entity $E_i \in \mathbb{E}$ and edges capture relative spatial features:

$$e_{ij} = \left[x_{ij}, y_{ij}, \frac{w_i}{h_i}, \frac{h_j}{h_i}, \frac{w_j}{h_i} \right]$$

Graph convolution is then applied over node-edge-node triplets using self-attention to propagate contextual information.

PICK (Yu et al. 2021) extends this design with multimodal node features, combining transformer-based text embeddings and CNN-based visual features. The edge representation is also enriched with sentence length information. To enhance tagging performance, PICK applies a soft adjacency matrix for graph learning, yielding task-specific node embeddings.

FormNet (Lee et al. 2022) introduces a novel attention mechanism—*Rich Attention*—that incorporates positional order and pixel distances (along x/y axes) into self-attention scores. This allows transformers to be aware of layout order and spatial proximity. Additionally, a graph encoder integrates neighborhood context prior to serialization, mitigating order ambiguity.

FormNetv2 (Lee et al. 2023) further enhances this by introducing visual cues as edge features and applying contrastive learning. Two types of graph corruption—topological (edge dropping) and feature-level (modality masking)—are used to generate augmented views. The model then learns robust multimodal embeddings using a normalized temperature-scaled cross-entropy loss (Sohn 2016).

In summary, these relation-aware models highlight the shift from flat sequence modeling to structure-aware document representation. While their specific graph construction and fusion techniques vary, they all aim to capture the rich spatial and semantic dependencies crucial for key information extraction and related tasks.

3.1.3 Few-shot learning frameworks

Extracting key information from VRDs using deep learning typically requires extensive manual annotation. However, in many real-world scenarios, obtaining large-scale, high-quality labels is costly and time-consuming. To address this challenge, few-shot and one-shot learning frameworks have been developed to enable key information extraction with minimal annotated data.

LASER (Wang and Shang 2022) builds upon the architecture of **LayoutReader** (Wang et al. 2021), reformulating entity recognition from a sequence labeling task into a generative one. By embedding entity type information—referred to as label surface names—into the target sequence, LASER enables the model to become label-semantic aware. It uses a *Partially Triangular* attention mask, allowing a single encoder to encode both the source text and the generated sequence. The input sequence $\{t_1, \dots, t_n\}$, comprising the sum of word, spatial, and positional embeddings, is fed into the encoder $\mathcal{E}$ with full self-attention. The decoder attends only to previously generated tokens. The generative target format is defined as:

$$t_{i-1}, [B], t_i, \dots, t_j, [E], e_1, \dots, e_k, [T], t_{j+1} \quad (1)$$

where $[B]$ and $[E]$ denote the entity span, e_i represent the label surface name, and $[T]$ marks the end of the label name. Since label tokens and special symbols do not appear in the source, learnable embeddings are assigned to them. A binary classifier predicts whether the next token should be copied from the source, effectively guiding the generative process.

Chen et al. (2023) propose an entity-level N -way *soft- K -shot* learning framework for key information extraction under few-shot settings. Targeting rare or unseen entity types, their approach employs meta-learning (Snell et al. 2017; Oreshkin et al. 2018) with a hierarchical decoder and contrastive learning module (**ContrastProtoNet**) for task personalization and improved adaptation. They also introduce **FewVEX**, a benchmark dataset designed for entity-level few-shot VRD extraction, and demonstrate that their method significantly outperforms existing meta-learning baselines in robustness and performance.

In the context of **one-shot** learning, Cheng et al. (2020) propose a graph-based method that transfers spatial relationships from support to query documents. Their approach leverages attention mechanisms to model interactions between static regions (keys/landmarks) and dynamic regions (values/fields), enabling accurate label probability distribution inference. A self-attention module captures dependencies among field entities, while a pairwise Conditional Random Field (CRF) with belief propagation completes the inference, resulting in an end-to-end trainable pipeline.

3.1.4 Prompt-learning frameworks

Prompt learning is a technique in natural language processing that guides models using specific prompts to elicit targeted responses. With the emergence of large-scale models, it has become an effective way to leverage contextual representations and the implicit knowledge embedded within pretrained models. Since most LLMs (OpenAI 2023; Touvron et al. 2023) and MLLMs (Liu et al. 2024) are primarily trained on plain text or natural images, layout-aware prompting and in-context learning methods (Brown et al. 2020) have been proposed to adapt these models for VRD-CU.

QueryForm (Wang et al. 2023) introduces a query-based framework for zero-shot document key information extraction. It employs a dual prompting mechanism—*entity prompts* (E-prompts) and *schema prompts* (S-prompts)—to transfer knowledge from large-scale, weakly annotated pretraining data (e.g., web pages) to target domains. The design ensures that the pretraining and fine-tuning objectives are well aligned, enabling consistent query-conditional predictions across both stages. During pretraining, HTML tags are used to construct the E-prompt e_p , while the S-prompt $\tilde{s}_p$ is derived from webpage domain information. In the fine-tuning stage, e_p becomes predefined, and s_p is treated as a set of learnable vectors. Let x denote the serialized text of the input document. The prediction targets during pretraining and fine-tuning are defined as:

$$\hat{y} = \mathcal{F}([s_p; \mathcal{E}[e_p, x]]), \quad (2)$$

$$y = \mathcal{F}([\mathcal{E}[\tilde{s}_p; e_p, x]]) \quad (3)$$

where $\mathcal{E}$ is the feature encoder and $\mathcal{F}$ represents the remaining layers of the language model. The training objective is to minimize the cross-entropy loss between $\hat{y}$ and y .

GenKIE (Cao et al. 2023) proposes an encoder–decoder-based multimodal KIE framework to leverage prompt to adapt various datasets and better leverage multimodal information. Following (Xu et al. 2020a, b), different encoding methods are adopted to acquire textual, layout and visual embeddings. Byte Pair Encoding is used as language pretrained backbones, and the OCR extracted document content is concatenated with predefined prompts split by the “[SEP]” token between OCR tokens and each prompt. The 2D-positional encoding introduced by Xu et al. (2020a) is used to acquire the layout embedding of each OCR extracted token, and ResNet (He et al. 2016) is used to extract the visual representations following Wang et al. (2022). The multimodal representations are fed into an encoder to learn interactively between modalities. Prompts, inserted at the end of the encoder’s textual inputs, are either template-style or question-style. For the entity extraction task, the prompt specifies the target entity type, and the decoder generates the entity value (e.g., for “Company is?”, the decoder outputs the company name). For the entity-labeling task, the prompt includes the value, and the decoder provides the entity type (e.g., for “Es Kopi Rupa is [SEP]”, the decoder identifies the entity type).

ICL-D3IE (He et al. 2023) is the first framework to employ LLMs with in-context learning to extract key information from VRDs using the iterative updated diverse demonstrates. Before designing the initial diverse demonstrations, the most similar n training documents to the n test samples need to be selected by calculating the cosine-similarity of document representations encoded by Sentence-BERT (Reimers and Gurevych 2019). Then, different types of demonstrations are introduced to integrate multiple-view context information into LLMs. Hard Demonstrations highlight the most challenging cases, are initially designed based on the incorrectly predicted cases from GPT-3 (Brown et al. 2020) predictions and are updated based on prediction results during the training process. Layout-aware demonstrations are created by selecting the adjacent hard segments to understand the positional relations. Formatting demonstrations are designed to guide LLMs in formatting the outputs for easy post-processing.

LMDX (Perot et al. 2023) designs a pipeline to use arbitrary LLMs to extract singular, repeated and hierarchical entities from VRDs. The document images are first fed into off-the-shelf OCR tools and are divided into smaller document chunks to be processed by the LLMs with accessible input length. Then, prompts are generated under XML-like tags to control the LLM’s responses and mitigate hallucination. Document Representation is a prompt contains the chunk content with the coordinates of OCR lines to bring layout modality to LLMs. After that, the task description and scheme representation prompts are designed to explain the task to accomplish and determine the output format. During inference, N prompts with K LLMs completions are generated to sample the correct answers.

3.1.5 Summary of key information extraction

Several models like Chargrid (Katti et al. 2018) and ACP (Palm et al. 2019) have enhanced VRD-CU by integrating visual and textual information. Additionally, auxiliary tasks such as OCR, utilized by Zhang et al. (2020); Wang et al. (2021), aid in improving multimodal feature representations through joint training. However, these frameworks often rely on smaller, randomly initialized models, which produce less representative features compared to those generated by large-scale pretrained models like LayoutLM (Xu et al. 2020a) and SelfDoc (Li et al. 2021a). Documents typically exhibit specific layouts and logical struc-

tures, which has prompted many models (Yu et al. 2021; Lee et al. 2022, 2023) to adopt graph-based approaches. These methods capture spatial and logical correlations among document elements, such as key-value pairs, leading to a more comprehensive document representation. While these frameworks have achieved improvements in document representation, their effectiveness hinges on having sufficient well-annotated training samples, which are time-consuming to acquire. This limitation has escalated the demand for few-shot (Wang and Shang 2022) and zero-shot (Wang et al. 2023) frameworks, which leverage contrastive learning and innovative attention mechanisms. Additionally, prompt learning has been applied to distill implicit knowledge from large-scale layout-aware pretrained models (Hong et al. 2021; Tu et al. 2023) and large language models (LLMs/MLLMs) (He et al. 2023; Perot et al. 2023). Despite these advances, a performance gap remains between well-fine-tuned models and few/zero-shot frameworks, highlighting the ongoing challenges in VRD-CU optimization.

3.2 Document entity linking

Documents are normally structured hierarchically, where parent-child relations always exist in various documents, such as key-value pairs in forms and section paragraphs in reports or papers. Unlike most VRD key information extraction models, which focus on recognizing the semantic entity categories in a sequence tagging task ignoring the relation between entities, linking the logical associations between document semantic entity pairs has recently been of greater interest.

3.2.1 Entity-level entity linking

Document entity linking aims to identify the relation between document entities. Some frameworks use the known entity bounding boxes, ignoring the entity recognition step and mainly focusing on exploring the relation between input document entities.

DocStruct (Wang et al. 2020) is the first entity linking framework that predicts hierarchical relationships between document semantic entities using multimodal features. It extracts the [CLS] token from a BERT-like encoder for textual representation (T_e) and uses an RNN to encode sequential RoI visual features (V_e) from ResNet-50. The layout feature (P_e), a linear projection of the bounding box coordinates $[x_1, y_1, \dots, x_4, y_4]$, is concatenated with T_e . The final entity representation is computed as:

$$\alpha = \text{Sigmoid}(W[T_e, P_e, V_e] + b), E = [T_e; P_e] + \alpha V_e, \quad (4)$$

where α gates the influence of visual features, following Wang et al. (2019). For each entity pair (E_i, E_j) , the probability of a parent-child link is $P_{i \rightarrow j} = E_i M E_j$, where M is an asymmetric parameter matrix. Negative sampling (Mikolov et al. 2013) is used during training to address sparsity and class imbalance.

SERA (Semantic Entity Relation extraction As dependency parsing) (Zhang et al. 2021) frames entity linking as a dependency parsing task. It uses *LayoutLM* (Xu et al. 2020a) to extract entity-level textual features T_e , concatenated with projected label embeddings L_e to form entity representations $E = [T_e, L_e]$. Contextual encoders (e.g., Transformer, BiLSTM, Graph) process E , and a Biaffine parser computes relation scores:

$$\begin{aligned} h_i^{key} &= \sigma(W^{key}E_i + b^{key}), \quad h_j^{value} = \sigma(W^{value}E_j + b^{value}), \\ p_b &= h_i^{key}W_{b1}h_j^{value} + h_i^{key}W_{b2}. \end{aligned} \quad (5)$$

To incorporate layout, a 2D layout feature l_{ij} (min width/height distance) produces a layout score $p_l = W_l l_{ij} + b_l$. The final relation score is $p = p_b + p_l$, used for binary or multi-label classification.

KVPFormer (Hu et al. 2023) reformulates entity linking as a QA task. It uses a Transformer-based encoder–decoder to model joint-grained representations. Each entity’s embedding $E = [T; l]$ is formed by averaging token-level features T and concatenating with its label embedding l . A Transformer encoder with spatial attention bias models entity relations. A binary classifier first detects key entities (questions), which are passed to a DETR-style decoder (Carion et al. 2020) to predict related entities (answers). For each key entity, top-K answer candidates are ranked by a sigmoid score and refined via a softmax-based coarse-to-fine strategy.

3.2.2 Word-level entity linking

As acquiring entity information needs prior knowledge from manual annotation or layout analyzing models, some works utilize serialized OCR-extracted sequence of words as inputs to extract the structured relations. However, as the logical relation links semantic entities, word-level frameworks must group word tokens into entities before exploring their association.

Carbonell et al. (2021) introduces a framework comprising three modules for sequentially conducting token grouping, entity labeling and relation prediction. Firstly, each text token t is represented by $[L_t; T_t]$ where $L_t = [x, y, w, h]$ is the coco format bounding box coordinates of t and T_t is the word representations. All tokens are fed into a token grouping GNN, $\mathcal{G}_{group}$, as a node where the edge between nodes is determined by k-NN to avoid high consumption of fully connected GNN. The $\mathcal{G}_{group}$ is trained on a link prediction task to predict the edge score between two nodes to group words, of which scores larger than a predefined threshold ρ . Then, the grouped words are fed into a Graph Attention Network (GAT) to use multi-head attention to aggregate words into entities and follow an MLP to conduct node classification to predict the category of each document entity. At last, another link prediction GNN, $\mathcal{G}_{link}$, is trained on edge classification based on aggregated entities.

SPADE (Hwang et al. 2021) formulate the word-level entity linking as a spatial dependency parsing task for serializing (ordering and grouping) tokens and predicting inter-group relation between grouped tokens. Firstly, a spatial text encoder is designed to make spatial-aware attention by introducing a relative spatial vector considering relative, physical and angle aspects. During this task, two binary matrices must be predicted M_g for token grouping and inter-group linking M_l . The vertices comprised by a entity types $\mathbb{V}$ and sequence of tokens $\mathbb{T}$ the encoded entity category and token are represented as c and t , respectively, and the relation score between vertices $v_i \rightarrow v_j$ can be calculated by:

$$h_i = \begin{cases} c_{v_i}, & \text{for } v_i \in \mathbb{V}, \\ W_h t_{v_i}, & \text{otherwise} \end{cases}, \quad (6)$$

$$d = W_d v_j, \quad (7)$$

$$s_0 = h_i^T W_0 d, \quad s_1 = h_i^T W_1 d. \quad (8)$$

The probability is acquired by $p_{ij} = \frac{\exp(s_{0,j})}{\exp(s_{0,j}) + \exp(s_{1,j})}$. An adjustable threshold is set to construct the M_g or M_l .

DocTR (Liao et al. 2023) formulate the entity linking as an anchor word-based entity detection and association problem. Each document entity is represented by the anchor word to convert the entity extraction and linking to word-level tasks. It contains a Deformable DETR-based vision encoder to extract multi-scale visual feature extraction. A **LayoutLM** based language encoder is applied to encode word-level textual representations. The outputs from vision/language encoders are fed into the vision-language decoder with the language-conditional queries to conduct entity extraction and linking. The decoder queries are one-to-one mapping with language encoder inputs.

3.2.3 Summary of entity linking models

Entity linking aims to identify logical connections between semantic entities within a document, typically formulated as a link prediction task. Early entity-level frameworks (Wang et al. 2020; Zhang et al. 2021) focus on learning multimodal representations of semantic entities and predicting parent–child relationships among document components. However, these approaches rely on preliminary entity-level bounding boxes, obtained via object detection models (He et al. 2017) or ground-truth annotations, which are often unavailable in real-world scenarios. To mitigate this, word-level frameworks (Carbonell et al. 2021; Hwang et al. 2021) jointly perform word grouping and entity linking, reducing reliance on bounding boxes during inference. Nonetheless, this increases token counts, potentially limiting scalability to longer documents. Moreover, joint-grained frameworks that integrate both fine-grained semantics and coarse-grained logical structures remain under-explored. Existing methods also predominantly target single-page documents with uniform layouts, neglecting the challenges posed by multi-page documents, complex form structures, and long-range dependencies.

3.3 VRD question answering

Unlike key information extraction, which targets specific details within document images, answering natural language questions involves interpreting more complex intentions and requires models to facilitate interactive understanding between the queries and document representations (Ding et al. 2022). The introduction of DocVQA dataset (Mathew et al. 2021) marked a significant shift in focus from natural scene images to text-dense, layout-aware single-page document images, establishing a benchmark in the field. As advancements have continued, demands have recently emerged for models capable of addressing more complex, multi-page scenarios (Tito et al. 2023; Ding et al. 2023, 2024). These emerging requirements highlight the need for models to process multimodal inputs and navigate through extensive documents, reflecting user inquiries' evolving complexity and naturalness in document-based question-answering systems. This section will briefly review the SoTAs in single-page document VQA models and introduce some recently proposed multi-page document understanding solutions (Tito et al. 2023; Blau et al. 2024).

3.3.1 Single-page VRD-QA

Similar to key information extraction, single-page question answering (QA) on VRDs often begins with classical pretrained language models such as BERT (Devlin 2018) and RoBERTa (Liu et al. 2019), which perform span-based QA by extracting relevant text token sequences. In addition, general-domain vision-language models like VisualBERT (Li et al. 2019), LXMERT (Tan and Bansal 2019), and ViLT (Kim et al. 2021) have been adopted to identify semantic entities in documents (Ding et al. 2023). Beyond these plain-text or general-domain visual-language models, numerous layout-aware architectures—specifically pretrained on document-centric data (see Sect. 4)—have achieved state-of-the-art performance on single-page document QA tasks. These approaches highlight the critical role of incorporating layout information during pre-processing to improve model performance on visually structured documents.

3.3.2 Multi-page VRD-QA

With the growing need to retrieve answers from multi-page documents (Tito et al. 2023; Ding et al. 2024), existing single-page models (Xu et al. 2020a; Wang et al. 2022; Hong et al. 2021) face limitations due to their 512-token input constraints. Recent advancements address this by introducing long-sequence transformers, hierarchical encoders, and page-locating modules tailored for multi-page VRD-QA.

Hi-VT5 (Tito et al. 2023) proposes a hierarchical encoder–decoder framework based on T5 (Raffel et al. 2020) for multi-page generative QA. Each page is encoded independently using a T5-based multimodal encoder, incorporating layout-aware question and OCR tokens (Biten et al. 2022), visual patch features from DIT (Li et al. 2022), and learnable page tokens. The enhanced page token representations (P') are aggregated and passed to the decoder to generate answers. As Hi-VT5 uses T5 (capable of handling up to 20,480 tokens), it significantly expands the input capacity beyond traditional VRD models. To compensate for T5's lack of inherent layout awareness, masked language modeling is employed during pretraining to inject visual and spatial signals.

GRAM (Blau et al. 2024) builds on a pretrained single-page model (DocFormerV2 (Appalaraju et al. 2024)) and introduces a lightweight global encoder between layers to enable inter-page interaction. Learnable page tokens attend across pages via sparse self-attention, capturing document-level context. To align with the pretrained backbone, ALiBi (Press et al. 2021) attention bias is applied to preserve attention to these new tokens. Unlike Hi-VT5, which only decodes from page tokens, GRAM incorporates all fine-grained token embeddings during decoding. It also employs C-Former (Raffel et al. 2020) to compress cross-page information, reducing computational overhead.

Kang et al. (2024) propose a two-stage framework using Pix2Struct (Lee et al. 2022) as a single-page encoder fine-tuned on DocVQA (Mathew et al. 2021). A self-attention-based scoring module is trained on top of the frozen encoder to select the most relevant page based on question-page matching. The selected page is then fed into the decoder to generate the answer, ensuring efficient retrieval by focusing only on the most informative content.

Overall, these multi-page VRD-QA frameworks advance beyond single-page limitations by leveraging hierarchical encoding, inter-page attention, and selective page routing to support more complex, large-scale document understanding tasks.

3.3.3 Summary of VRD-QA models

VRD-QA is a relatively new research area, introduced by DocVQA (Mathew et al. 2021), which focuses on answering natural language questions based on document images. Unlike key information extraction—which targets predefined key-value pairs—VRD-QA requires a deeper understanding of the entire document and its relevance to the question, demanding more comprehensive semantic and structural representations. Pretrained VRD-CU models such as LayoutLMv2 and LayoutLMv3 (Xu et al. 2020b; Huang et al. 2022) perform well on single-page document QA tasks. However, they face limitations when applied to multi-page documents due to input length constraints. To address this, recent works (Tito et al. 2023; Blau et al. 2024) have proposed identifying the most relevant page(s) and then applying single-page QA techniques to extract answers. Despite these advances, real-world applications often involve more complex scenarios—such as long-range dependencies and cross-page relationships—which remain open challenges in the VRD-QA field and warrant further investigation.

3.4 Summary of mono-task models

To achieve better performance on specific VRD-CU tasks, mono-task frameworks often adopt task-specific designs—for example, graph-based spatial encoding for KIE (Lee et al. 2022, 2023) or page-locating modules for multi-page VRD-QA (Tito et al. 2023). While optimized for individual objectives, these specialized components often form the foundation for multi-task learning. For instance, relation-aware modeling in KIE can support entity linking, and layout-aware encoders used in QA can be adapted for structured extraction. Recognizing these shared components is essential for understanding cross-task generalization and designing unified VRD-CU systems. Early frameworks relied heavily on hand-crafted features and rule-based heuristics, such as bounding box grouping or document-specific layout parsing. However, with the emergence of pretrained models like LayoutLM, BROS, and LiLT (Xu et al. 2020a; Hong et al. 2021; Wang et al. 2022), these manual designs have been largely replaced by shared multimodal encoders trained on large-scale document corpora. This shift has enabled the development of more general, instruction-tuned frameworks capable of handling multiple tasks with minimal adaptation. Section 4 builds on this trend, introducing multi-task models that unify these components through shared architectures, input granularity settings, and cross-task generalization strategies.

4 Multi-task VRD understanding models

As illustrated in Fig. 3, recent frameworks for VRD-CU increasingly adopt multi-task learning designs. These frameworks can be broadly categorized based on their **model architecture** and **input granularity**, which are key to determining their capability and generalizability. In terms of **architecture**, models are typically classified as **Encoder-only**, **Encoder-Decoder**. Encoder-only models utilize transformer-based encoders to capture multimodal document features and are often self-supervised pretrained on large-scale datasets. Encoder-Decoder models, usually generation-oriented, consist of an encoder that processes visual or textual input and a decoder that produces structured output. These can be

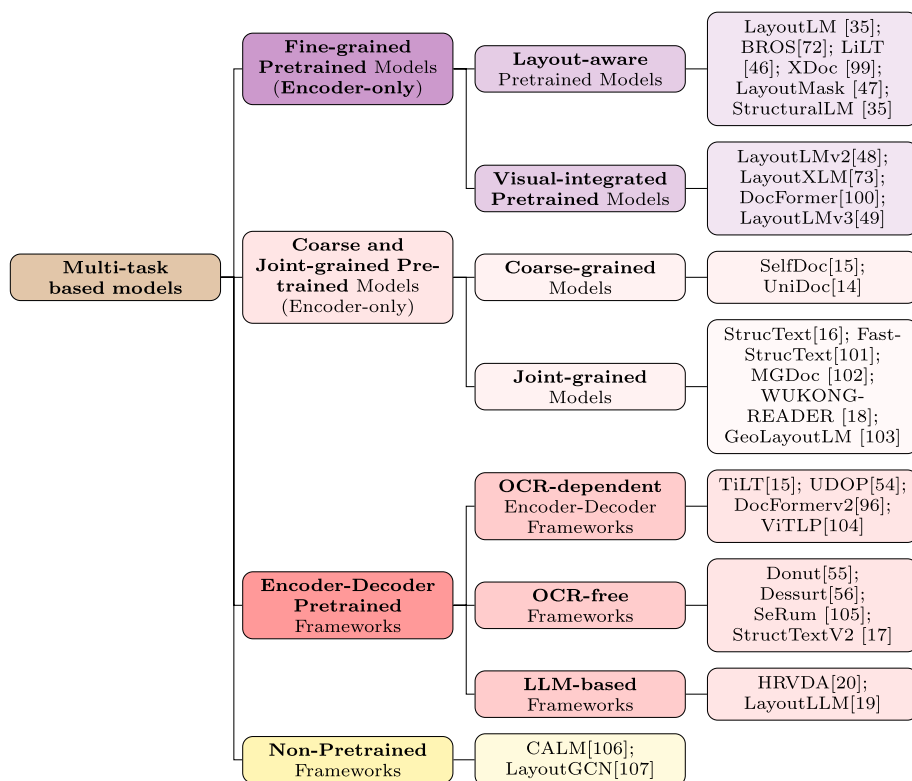


Fig. 3 Multi-task visually rich document understanding frameworks

further divided into **OCR-dependent** models, which require external text input, and **OCR-free** models that operate directly on document images. Non-pretrained models, in contrast, are task-specific and avoid extensive pretraining, often using lightweight or graph-based architectures. **Input granularity** distinguishes how models represent document information. **Fine-grained** models encode detailed features at the token or region level, preserving spatial and textual fidelity. **Coarse-grained** models abstract over larger structural units, such as blocks or paragraphs, to capture high-level semantics. **Joint-grained** approaches integrate both levels to support comprehensive document understanding.

4.1 Fine-grained pretrained models

Inspired by BERT-style pretrained models, many researchers have proposed effective methods to integrate layout and visual information into models, aiming to enhance the comprehensiveness of textual token representations.

4.1.1 Layout-aware pretrained language modeling

Understanding layout structure and spatial correlations between textual tokens can yield a more comprehensive representation of documents, going beyond what plain-text input

offers. To this end, various methods have been proposed to encode layout features. These methods, combined with tailored pretraining tasks, enable models to capture layout-aware information better and effectively fuse textual and layout features.

LayoutLM (Xu et al. 2020a) is the first pretrained document understanding model by leveraging textual and layout information in the pretraining stage. BERT architecture is the backbone and 2-D positional embedding¹ with textual information is used to pretrain on IIT-CDIP Test Collection 1.0. Two specific pretraining tasks are first introduced, named *Masked Visual-Language Model* (MVLM) and *Multi-label Document Classification*, to generate layout-aware textual representation and more comprehensive document representation, respectively. Like Masked Language Modeling adopted by most pretrained language models, MVLM randomly masks some input tokens but keeps the corresponding 2-D position embeddings to predict masked tokens to ensure the pretrained model is aware of the spatial relations between input tokens. MDC is a supervised pretraining task to predict the input document types (e.g. forms, exam papers, academic papers) that generate more comprehensive document-level representations. The fine-tuned LayoutLM could perform much better than textual-only frameworks on key information extraction (Jaume et al. 2019; Huang et al. 2019) and document classification (Harley et al. 2015).

BROS (Hong et al. 2021) proposes a pretrained VRDU model that captures the continuous nature of 2D space through a novel positional encoding and a correlation-aware attention mechanism. Given a token's bounding box coordinates $[x_1, y_1, \dots, x_4, y_4]$, the positional embedding is computed as:

$$\text{pos}_t = W_{p1}p_1 + W_{p2}p_2 + W_{p3}p_3 + W_{p4}p_4,$$

where each $p_i = \mathcal{F}_{\sin}(x_i) \oplus \mathcal{F}_{\sin}(y_i)$ and $W_{p*} \in \mathbb{R}^{2d_{\text{pos}} \times d}$.

To replace vanilla self-attention, BROS introduces an attention score α_{ij} that captures intra- and inter-modal interactions between textual and positional features:

$$\begin{aligned} \alpha_{ij} = & \underbrace{(W^q T_i)^\top (W^q T_j)}_{\text{text-text}} + \underbrace{(W^q T_i \circ W^{\text{pos}} \text{pos}_i)^\top (W^{\text{pos}} \text{pos}_j)}_{\text{text-pos}} \\ & + \underbrace{(W'^{\text{pos}} \text{pos}_i)^\top (W'^{\text{pos}} \text{pos}_j)}_{\text{pos-pos}}. \end{aligned} \quad (9)$$

Inspired by SpanBERT (Joshi et al. 2020), BROS also adopts an area-masked language modeling task, masking tokens within randomly sampled rectangular regions to enhance span-level understanding.

StructuralLM (Li et al. 2021) is the first VRDU model to leverage image patches (termed “cells”) to group tokens and perform patch-level pretraining. Built on BERT, it computes multimodal patch representations P and introduces two pretraining tasks: *Masked Visual-Language Modeling* (MVLM) and *Cell Position Classification* (CPC). Each patch is defined by a bounding box (x_0, y_0, x_1, y_1) and encoded using the 2-D positional encoding from LayoutLM. Tokens $\{t_1, t_2, \dots, t_n\}$ within a patch share this 2-D position. A token t_i is represented by: $t_i = T_i + \text{pos}_{t_i}^{2D} + \text{pos}_i^{1D}$, where T_i is the token embedding, $\text{pos}_{t_i}^{2D}$ is the patch-level layout embedding, and pos_i^{1D} is the token's sequence position. MVLM masks

¹Please refer to Sect. 6.1.3 for more detailed information on 2-D positional encoding.

tokens using patch-level layout embeddings, while CPC predicts the area index of a token among N equally divided regions. Together, these tasks encourage the model to capture spatial dependencies at the patch level.

LiLT (Wang et al. 2022) proposes a **language-independent** layout Transformer for mono- and multilingual document understanding. Text and layout are encoded separately and fused during pretraining, with their representations concatenated for downstream tasks. Each token t_j is represented as:

$$T_j = T_j + \text{pos}_{t_j}^{1D} + \text{pos}_{t_j}^{2D}.$$

Unlike LayoutLM, layout features are normalized to $[0, 1000]$, and six layout attributes $[x_0, y_0, x_1, y_1, w, h]$ are encoded via:

$$L = W_L(W_x x_0 \oplus W_y y_0 \oplus W_x x_1 \oplus W_y y_1 \oplus W_w w \oplus W_h h) + \text{pos}_L,$$

where $W_L \in \mathbb{R}^{6d'_L \times d_L}$ and W_* are learnable projections. Text and layout embeddings are processed via two sub-models, with a bi-directional attention complementation mechanism (BiACM) enhancing cross-modal interaction. Three pretraining tasks are introduced: MVLM, Key Point Location (predicting token area indices), and a text-layout alignment task (predicting whether text-layout pairs match). LiLT achieves strong results on multilingual document benchmarks (Wang et al. 2021; Xu et al. 2021).

XDoc (Chen et al. 2022) presents a unified architecture that handles diverse input formats—plain text, documents, and web data. It applies different encoding strategies: BERT-style for plain text; LayoutLM-style for documents with adapted 2D box encoding; and XPath-based embedding for web structures. All formats are pretrained with masked language modeling (MLM) and fine-tuned on corresponding benchmarks.

LayoutMask (Tu et al. 2023) targets the issue of improper reading order in OCR-dependent systems. It removes visual inputs and replaces global 1D positional encoding with a segment-based local variant, restarting the order within each OCR segment. The model introduces two pretraining objectives: (1) word-level MVLM with higher masking probabilities for segment boundaries, and (2) *Masked Position Modeling*, predicting the 2D coordinates of masked words to reinforce layout understanding.

4.1.2 Visual integrated models

Integrating visual cues with textual and layout information during pretraining significantly enhances a model's ability to capture rich and comprehensive document semantics. While earlier frameworks primarily focused on text and layout, recent approaches extend these by introducing visual-text alignment tasks to strengthen cross-modal understanding. This integration allows models to better interpret the intricate relationships among visual elements, textual content, and spatial structures within documents. Existing vision-integrated models can be broadly categorized based on how visual features are extracted: (1) *RoI-based models*, which employ region-of-interest (RoI) alignment using CNN backbones to generate high-level visual representations; and (2) *pixel-based models*, which directly extract pixel-level features via Vision Transformers.

LayoutLMv2 (Xu et al. 2020b) is the first pretrained model to integrate textual, layout, and visual modalities into a unified multimodal Transformer. Building upon LayoutLM, it introduces a trainable ResNeXt-FPN visual encoder whose outputs are projected to the same dimensional space as textual embeddings. Each modality is assigned a segment ID (e.g., seg_t , seg_v) added to its embeddings. Spatial-aware self-attention is adopted, incorporating 1D and 2D relative positional biases:

$$b^{1D} = W_{b_{1D}}(j - i), \quad (10)$$

$$b^{2D} = W_{b_x^{2D}}(x0_i - x0_j) + W_{b_y^{2D}}(y0_i - y0_j) \quad (11)$$

Three pretraining tasks are used: MVLM, *Text-Image Alignment* (TIA), and *Text-Image Matching* (TIM), enhancing cross-modal understanding.

LayoutXLM (Xu et al. 2021) extends LayoutLMv2 to multilingual settings with the same architecture but trains on 22 M multilingual PDF and 8 M scanned English documents. The MVLM task is adapted for multilingual contexts.

DocFormer (Appalaraju et al. 2021) proposes a multimodal encoder architecture combining LayoutLM-based text encoding and CNN-based visual encoding via ResNet-50. Textual ($\mathbb{T} \in \mathbb{R}^{d \times N}$) and visual ($\mathbb{V} \in \mathbb{R}^{d \times N}$) features are enhanced with 2D and 1D positional embeddings. A novel self-attention score α_{ij}^v includes multiple spatial and semantic biases:

$$\begin{aligned} \alpha_{ij}^v = & (W_v^K V_j)^\top (W_v^Q V_i) + (pos_{ij})^\top W_v^Q V_i + (pos_{ij})^\top W_v^K V_j \\ & + (W_s^Q pos_{ij}^{2d})^\top (W_s^K pos_{ij}^{2d}) \end{aligned} \quad (12)$$

Two modalities are fused at each encoder layer via addition: $\mathbb{M}_l = \mathbb{T}_l + \mathbb{V}_l$.

LayoutLMv3 (Huang et al. 2022) enhances model efficiency by eliminating heavy CNNs and adopting a pure Transformer-based vision encoder. Inspired by ViT, it divides document images into patches and projects them linearly with learnable 1D positional encoding. Layout information is encoded using segment-level bounding boxes with 2D positional embeddings. In addition to masked language modeling (MLM), LayoutLMv3 introduces two new pretraining tasks: *Masked Image Modeling* (MIM), where 40% of image patches are masked and reconstructed using discrete tokens with a cross-entropy loss (Bao et al. 2021); and *Word-Patch Alignment* (WPA), a binary classification objective that predicts whether a text token and image patch pair is aligned. Together, these tasks promote fine-grained multimodal alignment between visual and textual features.

4.2 Coarse and joint-grained pretrained models

Fine-grained models achieve state-of-the-art performance on many downstream tasks but face challenges with input length limitations and capturing document image layout and logical arrangement. To address these issues, coarse-grained or joint-grained frameworks have been introduced. To mitigate these limitations, these frameworks leverage multimodal information from document semantic entities such as paragraphs, tables, and textlines.

4.2.1 Coarse-grained frameworks

SelfDoc (Li et al. 2021a) is the first pretrained VRD-CU model leveraging coarse-grained document semantic entity for various understanding tasks. Unlike fine-grained OCR-based models, it uses Faster-RCNN to extract Regions of Interest (RoIs) from semantic entities, reducing input length and improving efficiency on dense, long documents. Visual embeddings are derived from RoIs, while Sentence-BERT (Reimers and Gurevych 2019) generates textual embeddings from the OCR-extracted text. These are fed into separate BERT-style encoders to model intra-modality context. A cross-modality encoder with cross-attention layers enables inter-modality learning. During pretraining, random masking is applied to either text or visual tokens. A modality-adaptive attention mechanism further adjusts the weight of visual and textual cues dynamically for robust entity-level representations.

UniDoc (Gu et al. 2021) is an coarse-grained model with a trainable image encoder using RoI-Align (He et al. 2017) to extract visual features and a novel cross-attention mechanism for multimodal fusion. It computes textual embeddings by combining average word embeddings and linearly projected bounding box coordinates. Product quantization (Jegou et al. 2010) is used to discretize RoI features into a finite visual codebook. A *Gated Cross-Attention* module is applied after multi-head fusion, where concatenated $[V : T]$ features are passed through a non-linear layer to generate modality-aware attention biases (β_v, β_t) . UniDoc employs three pretraining tasks: Masked Sentence Modeling, Visual Contrastive Learning, and Vision-Language Alignment (adapted from LayoutLMv2 (Xu et al. 2020b))—the last aligning image-text pairs at the entity level instead of region-level.

4.2.2 Joint-grained frameworks

StrucText (Li et al. 2021b) is a multimodal pretrained VRD-CU model that integrates fine-grained text and coarse-grained visual features to capture rich geometric and semantic information. Each text token is assigned a layout embedding $L = W_l[x_0, y_0, x_1, y_1, w, h]$ using $W_l \in \mathbb{R}^{6 \times N}$, and visual features are extracted from entities via a pretrained ResNet50-FPN. A segment-ID embedding aligns text and visual representations within the same entity. Three self-supervised tasks guide training: *Masked Visual-Language Modeling* (MVLN), *Segment Length Prediction* (SLP), and *Paired Box Direction* (PBD). SLP predicts entity lengths to encourage multimodal fusion, while PBD captures spatial relationships between entities.

Fast-StrucText (Zhai et al. 2023) builds upon StrucText to improve efficiency and representation power. It uses a more compact layout encoding $[x_0, y_0, x_1, y_1]$, an hourglass transformer with Merging and Extension blocks for token compression and recovery, and a *Symmetry Cross-Attention* (SCA) mechanism for enhanced modality interaction. Merging blocks apply weighted pooling to downsample token sequences, while extension blocks use repeat up-sampling. Fast-StrucText introduces new self-supervised tasks alongside MVLN: *Graph-based Token Relation* (GTR) for learning spatial relations, *Sentence Order Prediction* (SOP) for semantic continuity, and *Text-Image Alignment* to strengthen cross-modal grounding.

MGDoc (Wang et al. 2022) is the first multimodal, multi-granular pretrained framework designed to enhance inter-granular learning and cross-modal fusion through multi-granular and cross-modal attention mechanisms. It uses pretrained language encoders (Reimers and

Gurevych 2019) and vision backbones (He et al. 2016) to extract features across different levels—from word to page. These features are combined with positional embeddings (Zhai et al. 2023) and modality-type embeddings (Li et al. 2021b) to form the final input representations. To capture hierarchical relations among pages, entities, and words, two attention biases are added: a binary hierarchical bias indicating inclusion relations, and a relation bias based on relative bounding box positions. Cross-modal attention is used to integrate information across modalities. Pretraining involves three tasks: Masked Text Modeling (MTM), Masked Vision Modeling (MVM), and Multi-Granularity Modeling. MTM and MVM mask and reconstruct multi-granular representations using mean absolute error, while token-entity linking is modeled via dot-product similarity to promote multi-granular alignment.

WUKONG (Bai et al. 2022) utilizes fine-grained inputs (tokens with bounding boxes and document images) but enhances learning through coarse-grained self-supervised tasks. A Mask-RCNN backbone extracts visual features from textlines via RoI-head, while RoBERTa (Wei et al. 2020) encodes token representations in two stages: the first six layers for initial textual features (augmented with layout features (Xu et al. 2020b)), and the remaining layers for multimodal fusion with visual features. Four pretraining tasks are designed: Masked Language Modeling (MLM), Textline-Region Contrastive Learning (TRC), Masked Region Modeling (MRM), and Textline Grid Matching (TCM). TRC enhances entity-level cross-modal alignment via contrastive learning (Yao et al. 2021). MRM randomly masks 15% of textlines and predicts their visual embeddings (Li et al. 2021a). TCM divides the image into grids and predicts the grid location for masked textlines, strengthening layout awareness. Each task's loss is weighted with scaling parameters during training.

GeoLayoutLM (Luo et al. 2023) is a sophisticated multimodal framework that distinctively incorporates geometric information through specialized pretraining tasks and the development of innovative relation heads. Inspired by the dual-stream structure of **METER** and **SelfDoc** (Li et al. 2021a), GeoLayoutLM features separate vision and text-layout modules coupled with interactive co-attention layers that enhance the integration of visual and textual data. The model introduces two advanced relation heads—the Coarse Relation Prediction (CRP) head and the Relation Feature Enhancement (RFE) head—which refine relation feature representation crucial for both pretraining and fine-tuning phases. The pretraining regimen includes tasks designed to understand geometric relationships, such as GeoPair, GeoMPair, and GeoTriplet, aiding the model in grasping the complex dynamics of document layouts. During fine-tuning, the model utilizes pretrained parameters to optimize both semantic entity recognition and relation extraction tasks, employing a novel inference technique that enhances relation pair selection accuracy by focusing on the most probable relationships and minimizing variance among potential options.

4.3 Encoder–decoder pretrained frameworks

In addition to the above encoder-only frameworks, researchers have proposed encoder–decoder pretrained models that often approach tasks like Key Information Extraction (KIE) or Visual Question Answering (VQA) in a generative style. Addressing limitations of OCR-dependent frameworks, such as accumulated OCR errors and incorrect reading orders, OCR-free models have been introduced for end-to-end VRD-CU.

4.3.1 OCR-dependent encoder–decoder frameworks

TILT (Powalski et al. 2021) is a T5-based transformer encoder–decoder architecture enhanced with a relative spatial bias in the self-attention mechanism to acquire fine-grained token representations. It incorporates encoding methods that apply relative sequence input bias and capture horizontal and vertical distance biases in the attention scores. A U-Net-based framework is applied to extract the fixed-size feature maps fed into the encoder together. They follow T5 pretraining strategies on RVL-CIDP dataset (Harley et al. 2015) but use a salient span masking scheme adopted by Raffel et al. (2020); Guu et al. (2020). As the first encoder–decoder framework, **TILT** requires off-the-shelf OCR tools to acquire the textual token sequence. To conduct VRD content understanding end-to-end, some OCR-free frameworks are proposed to solve the limitations of OCR-dependent models.

UDOP (Tang et al. 2022) introduces an encoder–decoder framework for document understanding, built on a ViT-based model (Dosovitskiy et al. 2020) and inspired by LayoutLMv3 (Huang et al. 2022). It enhances textual token embeddings by summing them with aligned image patch embeddings when the token’s bounding box center overlaps with a patch. TILT-style positional biases are used, while ID embeddings are omitted. The decoder consists of two cross-attending modules: a bidirectional Transformer for text–layout decoding and an MAE-style vision decoder. Pretraining includes both self-supervised tasks (e.g., masked text–layout and image reconstruction) and supervised tasks using benchmark datasets. Additional objectives include layout modeling (predicting token group positions) and visual text recognition (identifying text at specific locations). Supervised fine-tuning is performed on datasets for classification (Harley et al. 2015), KIE (Stanisławek et al. 2021), VQA (Tanaka et al. 2021; Mathew et al. 2021), and layout analysis (Zhong et al. 2019).

DocFormerv2 (Appalaraju et al. 2024) is an encoder–decoder transformer architecture that uses multimodal (visual, textual and positional) to enhance the multimodal understanding and layout-aware language decoder to predict the predictions. The patched image pixels are fed into the convolutional and linear layers to get down-sampled patch embedding. The textual embeddings are acquired by linear projected token one-hot encoding. Both visual and textual embeddings are summed with the 2D-positional encoding (Xu et al. 2020a) of Patches and linear project bbox embedding ($[x_0, y_0, x_1, y_1]$) (Wang et al. 2022), respectively. Two encoder-based Token to Line (T2L), Token to Grid (T2G), and one decoder-based self-supervised learning task, MLM, are proposed to enable multimodal feature interactive learning. T2L aims to improve the relative position understanding between tokens by predicting the number of textlines between two randomly selected tokens. For improving the layout and structure, understanding needs to split the image into $m \times n$ grids to predict the located grid number of each OCR token. For the decoder MLM, the spatial feature of each masked text token is masked as well, and the other setup follows T5 (Raffel et al. 2020).

ViTLP (Mao et al. 2024) presents an encoder–decoder framework for OCR and document understanding. It employs a ViT-based vision encoder to extract image patch representations, which are decoded auto-regressively into text and layout sequences. A special “[LOC]” token encodes bounding box coordinates $[x_0, y_0, x_1, y_1]$ to reduce layout sequence length. To guide generation, “[BOS]” and “[CONT]” tokens are used—representing a token pair t_1, t_2 as “[BOS], t_1 , [LOC], t_2 , [LOC]”. The decoder has hierarchical heads: the text head predicts the next token using the full sequence, while the layout head predicts bound-

ing boxes only from “[LOC]” tokens. The “[CONT]” token enables variable-length generation by continuing decoding until “[END]”, based on the prefix token ratio.

4.3.2 OCR-free pretrained frameworks

Donut (Kim et al. 2022) is the first OCR-free VRD understanding model to understand and extract key information from input document images. Donut contains a Swin Transformer-based visual encoder to encode the input document image into image patches, which are then fed into a BART-based (Lewis et al. 2020) decoder pretrained on multi-lingual scenarios. During model training, teacher forcing is applied, and in the test stage, inspired by GPT-3 (Brown et al. 2020), prompts with special identify tokens are fed into the model for different downstream tasks. The output token sequence contains the special tokens $\langle START_* \rangle$ and $\langle END_* \rangle$ to identify the type of tasks and structure predict entities. The wrongly structured entity will be treated as an empty prediction. The model is pretrained on next-token prediction on the IIT-CDIP dataset and a Synthetic Dataset, which can be interpreted as a pseudo-OCR task. Similar to Donut, **Dessurt** (Davis et al. 2022) also proposes an encoder–decoder architecture but a different decoding process. Instead of using BART, the cross-attention used in Dessurt attends to all visual, query and previously generated textual information and is pretrained on more synthetic datasets with different font sizes and handwritten content.

SeRum (Cao et al. 2023) introduces an end-to-end OCR-free framework that decodes text based on local visual cues and spatial focus. A Swin Transformer extracts image patch features, which are refined through a Query Decoder to form vision-enhanced query representations, subsequently used by a Text Decoder for auto-regressive text generation. To better capture relevant regions, a Content-aware Token Merge module selects top- K visual tokens based on their correlation with the query, while unselected background tokens enhance the foreground via attention. Three pretraining tasks are proposed: Query-to-Segmentation (Q2S), which predicts instance masks for text regions; Text-to-Segmentation (T2S), which segments based on the decoder’s output; and Segmentation-to-Text (S2T), which generates text from segmented features. These tasks jointly enhance SeRum’s ability to detect, align, and generate text directly from document images.

StrucTextV2 (Yu et al. 2022) is an end-to-end structure that uses image-only input to conduct several downstream tasks. It contains a CNN-based visual extractor with FPN strategies (Lin et al. 2017) and follows ViT (Dosovitskiy et al. 2020) to get linear projected flattened patch-level representations. The patch token embeddings serve as the input to the Transformer encoder to enhance the contextually semantic representations. Then, the lightweight fusion network is applied to generate the final representations and fed into two branches during pretraining: made language Modeling (MLM) and Masked Image Modeling (MIM). Instead of using text inputs when MLM is used by other models (Devlin 2018), a portion of the text regions are masked with RGB values [255, 255, 255] randomly with a 2-layer MLP decoder to predict the masked token. MIM masks the rectangular text regions and predicts the RGB values of the missing pixels to improve the document representations. Except for the global average pooled FPN fused visual representations, the MLM-generated hidden state of each text region is concatenated and fed into a Fully Convolutional New York to get the regressed masked missing pixel values.

4.3.3 LLM-based frameworks

With the rapid development of LLMs/MLLMs, many frameworks are adopted to tackle document-based tasks. Both open-source (Bai et al. 2023; Liu et al. 2024) or close-source frameworks achieved promising performance on many benchmark datasets. This section introduces frameworks specifically proposed to address document related techniques with novel architectures. The trend of applying general domain LLMs/MLLMs in VRD-CU are discussed in Sect. 6.5.

DocLLM (Wang et al. 2024) proposes to address VRD-CU tasks by designing a disentangle spatial attention mechanism to capture alignments between text and layout modalities instead of integrating heavy visual backbones. The attention score between tokens i and j is computed as:

$$A_{i,j} = Q_i^t(K_j^t)^\top + \lambda_{t,s}Q_i^t(K_j^s)^\top + \lambda_{s,t}Q_i^s(K_j^t)^\top + \lambda_{s,s}Q_i^s(K_j^s)^\top \quad (13)$$

where Q^t , K^t and Q^s , K^s are projections from textual and spatial embeddings, respectively. Pretraining adopts an auto-regressive block infilling objective to better handle irregular layouts, where coherent OCR-derived blocks are masked and predicted given both prefix and suffix context. DocLLM is then instruction-tuned on 16 datasets covering four tasks (VQA, KIE, etc) using natural prompts, enabling strong generalization without vision encoders.

LapDoc (Lamott et al. 2024) enhances LLM-based VRD-CU framework by introducing a Layout-Aware Positional (LAP) embedding mechanism. It combines a learnable layout embedding e_l with the standard sinusoidal token positional encoding e_p , forming a unified input representation $x = e_t + e_p + e_l$, where e_t is the text embedding. The LAP encoder maps 2D bounding boxes into embeddings that capture spatial structure while preserving the auto-regressive training scheme. Pretraining is conducted using standard language modeling on 240 M document pages from the GLaMM corpus, integrating both text and layout features. Instruction tuning is performed on 9 datasets spanning four tasks—VQA, classification, KIE, and NLI—using task-specific prompt templates to guide model behavior, thereby enabling generalization across diverse document understanding tasks.

Some positional encoding strategies are proposed to enhance LLM/MLLM-based VRD-CU applications. **LayTokenLLM** introduces a layout token that summarizes the spatial configuration of each text segment using an attention-based layout tokenizer. This layout token is inserted alongside the text tokens and shares the same positional index as the first token in the segment, allowing the model to integrate layout cues without disrupting the sequential language modeling objective. By maintaining a full 100% T-Ratio (i.e., all positions contribute to text prediction), LayTokenLLM ensures efficient learning and inference, outperforming earlier layout-as-token baselines. In parallel, **Group Position Embedding (GPE)** proposes a complementary approach by assigning different spatial features—such as bounding box coordinates and reading order—to distinct attention head groups within rotary or sinusoidal positional encoding. This CoT-style decomposition enables the model to reason over multiple layout dimensions independently, enhancing spatial awareness without altering the model architecture. While LayTokenLLM focuses on embedding layout into the input sequence, GPE injects layout awareness through grouped attention, and both methods demonstrate strong performance across document benchmarks with minimal architectural overhead.

DocLayLLM incorporates explicit visual information by encoding image patches of document regions into external memory via a vision encoder (e.g., DiT), enabling stronger grounding in complex layouts. Crucially, it introduces Chain-of-Thought (CoT) prompting by fine-tuning on multi-step reasoning sequences, which guide the model to generate intermediate layout-aware explanations before predicting the final answer. This visual CoT-enhanced framework significantly boosts multi-hop document reasoning and better captures hierarchical and spatial relationships, outperforming prior text-only models on VQA, KIE, and NLI benchmarks.

LayoutLLM (Luo et al. 2024) introduces an LLM/MLLM-based approach integrated with a pretrained document understanding model to better fuse multimodal information. The input document's visual, textual, and layout information and any question text are encoded by a pretrained LayoutLMv3 (Huang et al. 2022) encoder and projected into the same embedding space as the adopted LLM, Vicuna-7B-v1.5 (Zheng et al. 2024). The method incorporates layout-aware pretraining tasks at three levels: document-level (e.g., document summarization), region-level (e.g., layout analysis), and segment-level (e.g., MVLM). These tasks enable the model to achieve comprehensive document understanding. Additionally, a novel module called LayoutCoT is designed to help LayoutLLM focus on question-relevant regions and generate accurate answers through intermediate steps. GPT-3.5-turbo (OpenAI 2023) is used to prepare the dataset for document summarization training and to construct LayoutCoT training data.

HRVDA (Liu et al. 2024) aims to propose a MLLM accepting high-resolution image inputs to conduct fine-grained information extraction from VRDs. A swin-transformer (Liu et al. 2021) is used to encode document images into image patch tokens. A pluggable content detector then identifies visual tokens that contain relevant document content information. Following this, a content filtering mechanism performs token pruning to remove irrelevant tokens. The remaining encoded visual tokens are processed through an MLP to ensure consistency with the LLM embedding space dimensions. These pruned tokens are then fused with instruction features, allowing further filtering of tokens irrelevant to the instructions. The final streamlined set of visual tokens and instructions is fed into the LLM, which generates the corresponding responses.

4.4 Non-pretrained frameworks

CALM (Du et al. 2022) introduces a common-sense augment document understanding framework to understand the query and extrapolate answers not contained in the context of the input document image. They follow LayoutLMv2 (Xu et al. 2020b) to encode input document multimodal representations. The textual token embeddings are fed into a Document Purifier component to merge the tokens $\{t_1, \dots, t_n\}$ belonging to one entity type N to one Upper Layer token $\hat{c}$ by applying average pooling of $\hat{c} = AvePool(t_1, \dots, t_n)$. Each Upper Layer token is concatenated with the commence augmented on ConceptNet NumberBatch (Speer et al. 2017) entity word vector c' to get the final entity representation $c = concat(\hat{c}, c')$. A similar Question-Purifier is applied to use common-sense knowledge to enhance the question representation. Then, with the assistance of ConceptNet, relevant common-sense knowledge is recalled based on the common-sense representation of both documents and queries. By considering the predicted question-answer relationship, a final

self-attentive graph convolutional network following (Zhu et al. 2021) is proposed to address document reasoning tasks more effectively.

LayoutGCN (Shi et al. 2023) proposes a lightweight and effective model which contains a fully connected graph where text blocks are nodes and edges connect every two blocks. The model architecture includes a TextCNN-based (Kim 2014) encoder to encode N-gram textual embeddings, a linear trainable layout encoder to project the normalized bbox coordinates into hyperspace following other layout-aware models, and a visual encoder (CSP-Darknet Wang et al. (2020) for document image features). These features are integrated using a Graph Convolution Network (GCN) to capture relationships between nodes. The final node representation combines text, layout, and visual information, benefiting various VRD-CU tasks.

XYLayoutLM (Gu et al. 2022) builds on **LayoutXLM** (Xu et al. 2021) by introducing two key modules: an Augmented XY Cut and a Dilated Conditional Position Encoding (DCPE). The Augmented XY Cut improves traditional XY Cut (Nagy and Seth 1984) by incorporating adjustable thresholds (λ_x, λ_y) and a shift factor (θ) to refine token clustering and correct OCR-induced reading order errors through recursive XY Tree-based segmentation. DCPE addresses the limitations of Conditional Position Encoding (CPE) (Chu et al. 2022) by separately modeling textual and visual modalities. It uses 1D convolutions for textual tokens to capture local sequence structure and dilated convolutions (Yu and Koltun 2015) to encode long-range dependencies without added complexity, improving multimodal representation learning in document understanding tasks.

4.5 Summary of multi-task frameworks

Various models are proposed to enhance document representations for VRD-CU tasks by leveraging pretrained language models to enrich text token sequences with layout information through positional encoding (Xu et al. 2020a), attention mechanisms (Wang et al. 2022), and layout-aware tasks (Tu et al. 2023). However, VRDs contain rich visual details like font, texture, and color and visually complex entities such as tables, charts, and photos. Many models (Xu et al. 2020b; Huang et al. 2022; Appalaraju et al. 2021) integrate visual cues to enhance fine-grained document features, but their quadratic time and space complexity pose challenges for handling long sequences in multi-page document understanding (Ding et al. 2024). Fine-grained models excel but struggle with capturing layout and structural details from document images. Coarse-grained frameworks (Li et al. 2021a; Gu et al. 2021) mitigate fine-grained limitations by leveraging entity-level multimodal information, yet compressing diverse entity aspects into a single dense vector risks losing information (Ding et al. 2024). Joint-grained frameworks (Li et al. 2021b; Zhai et al. 2023; Wang et al. 2022; Bai et al. 2022; Luo et al. 2023) integrate multi-grained information to produce comprehensive representations. Non-pretrained models leverage external knowledge (Du et al. 2022) or lightweight networks (Shi et al. 2023) to rival large-scale pretrained frameworks' performance. Most document understanding models (Xu et al. 2020a; Huang et al. 2022; Wang et al. 2022; Li et al. 2021a) rely on off-the-shelf OCR tools for text extraction, which can be susceptible to OCR quality issues and incorrect reading orders. OCR-free frameworks directly process document images to mitigate these limitations; However, these frameworks may exhibit sub-optimal performance compared to methods using established OCR tools with additional resource consumption.

5 Visually rich document content understanding datasets

Based on the downstream tasks differences, we summarize the Key Information Extraction with Entity Linking datasets and Visually Rich Document Question Answering dataset in Sects. 5.1 and 5.2.

5.1 Key information extraction and entity linking

5.1.1 Scanned receipt datasets

SROIE (Huang et al. 2019) is a widely used dataset for text localization, OCR, and key information extraction from scanned receipts, introduced in the ICDAR 2019 Challenge on “*Scanned Receipts OCR and Key Information Extraction*” The Key Information Extraction (KIE) task focuses on four key types: *Address*, *Date*, *Company*, and *Total*, with corresponding values provided in the annotation file for each receipt. The F1 score for this task is calculated based on Mean Average Precision (MAP) and recall. Note that entity-level annotations are not provided in the official dataset, requiring the use of external tools to obtain the information for coarse-grained or joint-grained models.

Payment-Receipts (Majumder et al. 2020) is a subset of SROIE, created by sampling up to 5 documents from each template in the original SROIE dataset. The template of each receipt is decided by the Company annotation. The target schema focuses on extracting only *Date* and *Total*. This subset is used to evaluate the model’s ability to handle unseen templates.

CORD (Park et al. 2019) is a widely used dataset for post-OCR receipt understanding, featuring two-level labels annotated by crowd-sourcing workers. It includes eight super-classes, such as *Store*, *Payment*, *Menu*, *Subtotal*, and *Total*, each with several subclasses. For example, *Store* contains subclasses like *Name*, *Address*, and *Telephone*. CORD provides both textline-level and word-level annotations for both fine-grained and coarse-grained frameworks, with some sensitive information blurred. All models are evaluated on the released first 1,000 samples.

5.1.2 Form-style datasets

FUNSD (Jaume et al. 2019) is derived from the RVL-CDIP dataset (Harley et al. 2015) by manually selecting 199 readable and diverse template form images. The dataset is annotated using the GuiZero library to provide both entity and word-level annotations, including manual text recognition. Semantic links indicate relationships between entities, such as Question-Answer or Header-Question pairs. Consequently, FUNSD supports key information extraction, OCR, and entity linking tasks.

XFUND (Xu et al. 2021) is the first multilingual dataset following the FUNSD format. It collects form templates in seven languages (Chinese, Japanese, Spanish, French, Italian, German, and Portuguese) from the internet. Human annotators fill these templates with synthetic information by typing or handwriting, ensuring each template is used only once. The filled forms are then scanned into document images, processed with OCR, and annotated with key-value pairs. Each language has 199 annotated forms, supporting multilingual key information extraction and entity linking tasks.

Payment-Invoice (Majumder et al. 2020) contains two corpora of invoices from different sources. The first corpus, Invoice 1, includes 14,273 invoices from various vendors with different template styles, used for training and validation. The second corpus, Invoice 2, comprises 595 documents with distinct templates not found in Invoice 1, serving as the test set. Human annotators extract six required keys from each single-page invoice, such as *Invoice Date*, *Total Amount*, and *Tax Amount*. This dataset is suitable for evaluating generative-style models and fine-grained sequence labeling models. For coarse-grained models, additional text line or entity-level information can be extracted using off-the-shelf tools.

VRDU-Registration Form (Wang et al. 2023) is a dataset of registration forms about foreign agents registering with the US government collected from the Federal Communications Commission. Commercial OCR tools extract the text content of the forms. Annotators draw bounding boxes around six unrepeatable entities (each entity appears only once per document) per document: *File Date*, *Foreign Principal Name*, *Registrant Name*, *Registration ID*, *Signer Name*, and *Signer Title*. The dataset provides entity-level annotations, which can be easily preprocessed to acquire word-level annotations, supporting any granularity of Key Information Extraction (KIE) models.

VRDU-Ad-buy Form (Wang et al. 2023) consists of 641 invoices or receipts signed between TV stations and campaign groups for political advertisements. It follows the same annotation procedure as the VRDU-Registration Form but involves a more complex schema. This includes nine unique entities (e.g., *Advertiser*, *Agency*, *Contract ID*), four repeated entities (e.g., *Item Description*, *Sub Prices*), and hierarchical entities (e.g., *Line Item*). Repeated entities may contain different values within a single document, while hierarchical entities comprise several repeated entities as components.

Form-NLU (Ding et al. 2023) is a visual-linguistics dataset designed to support researchers in interpreting specific designer intentions among various types of noise from different form carriers, including digital, printed, and handwritten forms. Fine-grained key-value pairs, such as *Company Name*, *Previous Notice Date*, and *Previous Shares*, are manually annotated. The training and validation set comprises 535 digital-born forms, with 76 reserved for validation. Additionally, three test sets are provided, containing 146 digital, 50 printed, and 50 handwritten form images, respectively. Form-NLU can be used to evaluate form layout analysis and Key Information Extraction (KIE) models of any granularity. With proper processing, it can also be used to evaluate entity linking frameworks, thanks to the well-annotated key-value pairs.

5.1.3 Multi-page datasets

Kleister-NDA (Stanisławek et al. 2021) is a dataset collected from the Electronic Data Gathering, Analysis, and Retrieval System (EDGAR) focusing on Non-disclosure Agreements (NDAs). During preprocessing, the collected 540 HTML files are converted into digital multi-page PDF files (totaling 3,229 pages) using the Puppeteer library. Four key items, *Effective Date*, *Party*, *Jurisdiction*, and *Term*, are manually annotated by three annotators to extract the corresponding values. The NDA dataset is widely used by fine-grained level models but may require additional processing for frameworks with limited sequence length due to the multi-page inputs.

Kleister-Charity (Stanisławek et al. 2021) contains 2778 annual financial reports from the Charity Commission, which lack strict formatting rules. The Charity Commission web-

site provides eight key pieces of information, such as Postcode, Charity Name, and Report Date. Annotators manually correct minor errors to ensure accuracy. Compared to Kleister-NDA, the Charity dataset has longer document inputs, totaling 61,643 pages, requiring models to handle long sequence outputs. Both Charity and NDA datasets provide only key-value pair annotations, making them suitable for the generation and fine-grained sequence labeling tasks but requiring additional processing to acquire entity-level annotations.

DocILE (Šimsa et al. 2023) comprises three subsets: an annotated set of 6680 real business documents, an unlabeled set of 932,000 real business documents for unsupervised pretraining, and a synthetic set of 100,000 documents generated with full task labels. Documents come from public sources like the UCSF Industry Documents Library and Public Inspection Files, with annotations for Key Information Localization and Extraction and Line Item Recognition. Synthetic documents were created using annotated templates and a rule-based synthesizer. DocILE provides entity-level annotations that can be easily post-processed to acquire word-level annotations.

5.1.4 Summary of KIE datasets

KIE datasets for VRDs have progressively expanded to encompass a broader range of domains, formats, and complexities, as shown in Table 1. Early benchmarks such as FUNSD (Jaume et al. 2019) and SROIE (Huang et al. 2019) primarily consist of single-page, printed documents with a limited set of predefined keys, featuring simplified annotation schema that facilitate model prototyping and benchmarking. In contrast, recent datasets like Kleister-NDA, Kleister-Charity (Stanislawek et al. 2021), and DocILE (Šimsa et al. 2023) incorporate multi-page documents reflective of real-world enterprise scenarios (e.g., contracts, invoices). These introduce challenges such as long context understanding, cross-page context modeling, hierarchical key structures. Further complexity arises from the inclusion of handwritten text (e.g., FUNSD, EPHOIE) and multilingual content (e.g., XFUND), necessitating models capable of handling OCR noise, variable layouts, and linguistic diversity. These trends reflect a shift toward more realistic KIE challenges, with future directions emphasizing multimodal reasoning, cross-lingual adaptability, and robust processing of complex, noisy document structures.

5.2 Visually rich document question answering

5.2.1 Single page VRD-QA datasets

DocVQA (Mathew et al. 2021) is a pioneering dataset in document-based Visual Question Answering (VQA), sourced from the UCSF Industry Document Library. It comprises 50,000 manually generated questions framed on 12,767 document images, encompassing digital, printed, and handwritten formats. The dataset follows an extractive-style QA format similar to benchmarks like SQuAD (Rajpurkar et al. 2016) and VQA (Biten et al. 2019a). Evaluation typically involves fine-grained models such as LayoutLM variants (Xu et al. 2020a, b; Huang et al. 2022), LiLT (Wang et al. 2022), and generative models (Tang et al. 2022; Kim et al. 2022), using metrics like Average Normalized Levenshtein Similarity (ANLS) (Biten et al. 2019b). However, it requires additional processing for coarse-grained models like SelfDoc (Li et al. 2021a). *CS-DVQA* (Du et al. 2022) builds upon DocVQA by

Table 1 Summary of key information extraction (KIE) datasets for VRDs

Name	Venue	Year	Domain	# Docs	# Images	# Keys	MP.	Language	Metrics	Format
FUNSD	ICDAR-w	2019	Multi-source	N/A	199	4	N	English	F1	P./H
SROIE	ICDAR-c	2019	Scanned Receipts	N/A	973	4	N	English	F1	P
CORD	Neurips-w	2019	Scanned Receipts	N/A	1,000	54	N	English	F1	P
Payment-Invoice	ACL	2020	Invoice Form	N/A	14,237+595	7	N	English	F1	D
Payment-Receipts	ACL	2020	Scanned Receipts	N/A	478	2	N	English	F1	P
Kleister-NDA	ICDAR	2021	Private Agreements	540	3229	4	Y	English	F1	D
Kleister-Charity	ICDAR	2021	AFR	2778	61,643	8	Y	English	F1	D./P
EPHOIE	AAAI	2021	Exam Paper	N/A	1494	10	N	Chinese	F1	P./H
XFUND	ACL	2022	Synthetic Forms	N/A	1393	4	N	Multilingual	F1	D./P./H
Form-NLU	SIGIR	2023	Financial Form	N/A	857	12	N	English	F1	D./P./H
VRDU-Regist. Form	KDD	2023	Registration Form	N/A	1915	6	N	English	F1	D
VRDU-Ad-buy Form	KDD	2023	Political Invoice Form	N/A	641	9+1(5)	N	English	F1	D./P
DocILE	ICDAR	2023	Invoice Form	6680	106,680	55	Y	English	AP, CLEval	D./P

MP.: Multi-page; F1: F1-score; P.: Plain-text; H.: Handwritten; D.: Digital

enhancing QA pairs to better reflect real-world requirements. It extracts 600 images from the DocVQA dataset and generates 1,000 QA pairs under human supervision. During question generation, it incorporates common-sense knowledge from real life, expanding answers beyond extractive in-line text to include question-related nodes (Nodes) sourced from ConceptNet (Speer et al. 2017).

VisualMRC (Tanaka et al. 2021) is compiled from website screenshots across 35 domains, carefully selected to exclude pages with handwritten content and to prefer pages containing short text (no more than 2 to 3 paragraphs). Unlike other datasets that might only provide question-answer annotations (Mathew et al. 2021) or automatically acquire document semantic entities (Ding et al. 2023), VisualMRC includes manually annotated layout structures with fine-grained semantic entity types such as *Heading*, *Paragraph*, *Subtitle*, *Picture*, and *Caption*. Question-answer pairs are generated through crowdsourcing. Consequently, VisualMRC is well-suited for evaluating both fine-grained and coarse-grained-

based QA frameworks, providing a rich resource for assessing the effectiveness of models in understanding and interpreting detailed document layouts and semantic entities.

PDFVQA-Task A and Task B (Ding et al. 2023) form part of the first document VQA dataset from PubMed Central, focusing on content and structural understanding. This dataset includes three tasks: two for single-page documents (Tasks A and B) and one for multi-page documents (Task C). Task A evaluates the structural and spatial relationships within document images, with answers being either counts or Yes/No. Task B focuses on extracting document entities based on their logical and spatial configurations. The PDFVQA dataset provides only coarse-grained, entity-level annotations, necessitating further processing for models that require fine-grained analysis. This setup is ideal for testing models' capabilities in understanding the logical and spatial structures of document images.

5.2.2 Multi-page VRD-QA datasets

TAT-DQA (Zhu et al. 2022), an extension of the TAT-QA (Zhu et al. 2021) dataset, is developed with more complex natural document structures and an expanded set of manually corrected and generated question-answer pairs derived from business financial reports. Unlike other datasets that primarily focus on extractive or simple abstractive answers (such as counting or yes/no), TAT-DQA includes questions requiring arithmetic reasoning, where values must be extracted from tabular data and textual content for discrete calculations. This dataset adopts the evaluation metrics of TAT-QA, including Exact Matching and a numeracy-focused F1 score. These metrics are particularly tailored to assess the accuracy of arithmetic reasoning and data extraction capabilities of the models tested with TAT-DQA.

RDVQA dataset (Wu et al. 2022) compiles a large collection of conversational chats and associated images from an E-commerce platform. It employs standard OCR and Named Entity Recognition (NER) techniques to extract text and redact sensitive information, ensuring privacy protection through masking. The dataset includes question-answer pairs within the images, which are manually verified to confirm image clarity and the presence of at least one question-answer pair per image. Although some documents span multiple pages, this dataset is structured such that it can be processed relatively easily by single-page VRD-QA models.

PDFVQA-Task C (Ding et al. 2023) is a distinct sub-task within the PDFVQA dataset that expands document VQA to encompass entire long documents, moving beyond the single-page focus of Tasks A and B. In Task C, to answer questions, the model often needs to retrieve information from multiple document entities. Thus, Task C employs Exact Matching for its ground truth annotations. Similar to Tasks A and B, additional processing is required for evaluating models at a fine-grained level.

MP-DocVQA (Tito et al. 2023) extends the original DocVQA (Mathew et al. 2021) dataset to accommodate multi-page document analysis. This version includes adjacent pages from the same documents, expanding the dataset from 12,767 to 64,057 document images. In adapting to a multi-page format, some questions inappropriate were removed. However, it's important to note that while the dataset allows for questions across multiple pages, the answers remain confined to individual pages; there are no cross-page answers in the MP-DocVQA dataset.

DUDE (Van Landeghem et al. 2023) is the first cross-domain, multi-page document VQA dataset, featuring a diverse collection of documents from various fields such as medi-

cal, legal, technical, and financial, and different document types including CVs, reports, and papers. It comprises 5019 documents, 28,709 document pages, and 41,541 manually annotated questions. Question types vary from extractive in-line text and Yes/No answers to multi-hop reasoning and structural understanding, similar to those in PDFVQA (Ding et al. 2023). To evaluate model performance, DUDE uses the ANLS metric (Mathew et al. 2021) for assessing answer prediction accuracy. Additionally, it employs two other metrics: Expected Calibration Error (Guo et al. 2017) and Area-Under-Risk-Coverage-Curve (CURC) (Geifman and El-Yaniv 2017; Jaeger et al. 2022) to gauge the overconfidence and miscalibration in document understanding models. These features make DUDE a comprehensive tool for evaluating cross-domain document understanding models.

MMVQA (Ding et al. 2024) is a dataset sourced from PubMed Central, designed for the retrieval of multimodal semantic entities from multi-page documents. The questions are generated using ChatGPT (OpenAI 2023) and subsequently verified manually. Unlike other datasets that focus solely on in-line text or text-dense entities, MMVQA also considers entire tables and figures as potential answers to the given questions. This dataset introduces various evaluation metrics to cater to different application scenarios: Exact Matching and Partial Matching Accuracy assess the precision of responses, while Multi-label Recall evaluates how well the model identifies all relevant answers across the document. This diverse set of metrics makes MMVQA suitable for comprehensive performance evaluation in complex, multimodal document understanding tasks.

5.2.3 Summary of VRD-QA datasets

Similar to KIE datasets, in Table 2, the development of VRD-QA datasets also demonstrates a marked progression toward practical and challenging scenarios. While earlier datasets (Mathew et al. 2021; Tanaka et al. 2021) concentrated on single-page, clean digital documents with simple textual answers, newer benchmarks increasingly demand complex reasoning over multi-page layouts and multimodal semantic entities (e.g., MPDocVQA Tito et al. (2023), MMVQA Ding et al. (2024)), accommodate diverse and domain-specific content (e.g., financial, academic, cross-domain), and support mixed formats including printed and handwritten inputs. The variety in answer formats—ranging from span-based text to entities, binary decisions, and reasoning-based generation—combined with rich evaluation protocols (e.g., EM, F1, ANLS, BLEU), underscores the rising complexity and real-world applicability of the task, pushing models toward deeper semantic comprehension, robust layout interpretation, and multimodal reasoning under noisy and heterogeneous conditions.

5.3 Evaluation metrics

F1-score The F1-score is a widely used evaluation metric in KIE, VQA, entity recognition, and document classification. It helps assess how well a model can identify relevant elements while minimizing false positives and false negatives. It provides a balanced measure of a model's precision and recall, calculated as the harmonic mean of precision and recall using the following formula:

$$F1 \text{ score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Table 2 Summary of visually rich document question answering (VRD-QA) datasets

Name	Venue	Year	Domain	# Doc	# Img	# Q	Answer type	MP	Format	Metrics	Anno
DocVQA	WACV	2021	Industrial reports	N/A	12,767	50,000	Text	N	D./P./H	ANLS	Human
VisualMRC	AAAI	2021	Website	N/A	10,197	30,562	Text	N	D	BLUE, etc	Human
TAT-DQA	MM	2022	Financial reports	2758	3067	16,558	Text/RS-Gen	Y	D	EM, F1	Human
RDVQA	MM	2022	Data analysis report	8362	8514	41,378	Text	N	D	ANLS, ACC	Human
CS-DVQA	MM	2022	Industry documents	N/A	600	1000	Text and Nodes	N	D./P./H	ANLS	Human
PD-FVQA-Task A	PKDD	2023	Academic paper	N/A	12,337	81,085	Num or Yes/No	N	D	F1	Template
PD-FVQA-Task B	PKDD	2023	Academic paper	N/A	12,337	53,872	Entity	N	D	F1	Template
PD-FVQA-Task C	PKDD	2023	Academic paper	1147	12,337	5653	Entity	Y	D	EM	Template
MP-DocVQA	PR	2023	Industrial reports	6000	48,000	46,000	Text	Y	D./P./H	ANLS	Human
DUDE	ICCV	2023	Cross-domain	5019	28,709	41,541	Text, Yes/No	Y	D	ANLS	Human
MMVQA	IJCAI	2024	Academic paper	3146	30,239	262,928	Entity	Y	D	EM, PM, MR	LLM + H

MP: Multi-page; D./P./H.: Digital/Printed/Handwritten formats; RS-Gen.: Reasoning-based Generation; Metrics: ANLS = Average Normalized Levenshtein Similarity, EM = Exact Match, F1 = F1-score, PM = Partial Match, MR = Multi-label Recall, ACC = Accuracy, BLEU = BLEU score; Anno.: Annotation source (Human, Template, or LLM + Human)

5.3.1 Evaluation metrics in KIE datasets

AP (Average Precision) AP measures the area under the precision–recall (P–R) curve for ranked retrieval results, calculating the average of precision values at different recall levels. In document understanding and object detection tasks, AP measures how accurately a model can identify and localize different document elements such as tables, figures, and text blocks.

CLEval (Character-Level Evaluation) CLEval (Baek et al. 2020) is designed for text detection and recognition tasks, providing a fine-grained assessment through: instance matching process and character scoring process. The instance matching process handles granularity differences between predicted and ground-truth text regions, and the character-

level scoring process measures partial correctness. CLEval evaluation metrics provides more nuanced evaluation by considering partially correct results.

5.3.2 Evaluation metrics in VRD-QA datasets

ANLS (Average Normalized Levenshtein Similarity) ANLS (Biten et al. 2019) is a metric designed to evaluate the similarity between predicted and ground truth answers, particularly in tasks such as VQA, where outputs may contain minor errors such as typos. Formally, ANLS can be defined as: For N questions, each with M ground truth answers a_{ij} and a prediction o_{q_i}

$$\text{ANLS} = \frac{1}{N} \sum_{i=1}^N \left(\max_j s(a_{ij}, o_{q_i}) \right) \quad (14)$$

where $s(a_{ij}, o_{q_i})$ is the normalized similarity for the j -th ground-truth to the i -th prediction. ANLS normalizes Levenshtein distance to account for differences in string lengths, producing a score between 0 and 1, where 1 indicates perfect similarity. ANLS gives a high score when predicted text is very similar to the ground truth and gracefully degrades as errors increase.

BLEU (Bilingual Evaluation Understudy) The BLEU score is a metric originally designed for evaluating machine translation quality, but has been adapted for various text generation tasks, including VQA. BLEU measures the similarity between machine-generated text and reference texts by comparing n -grams. BLEU scores range from 0 to 1, with higher scores indicating greater similarity to reference texts.

ACC (Accuracy) Accuracy measures the proportion of correctly predicted instances over the total number of instances. In VRD-QA tasks (Mathew et al. 2021), this typically refers to the percentage of questions where the predicted answer exactly matches at least one ground-truth answer. A stricter variant, *Exact Match (EM)*, scores an instance as correct only if the predicted entity set is identical to the ground-truth set (Ding et al. 2024). Conversely, *Partial Match (PM)* is a more lenient metric that gives credit for partial correctness—an instance is considered correct if there is any non-empty overlap between the predicted and ground-truth entity sets. While EM emphasizes exact retrieval, PM is more suitable for scenarios where partial information suffices, such as identifying at least one key entity within a document.

MR (Multi-label Recall) MR is a metric designed for multi-label classification tasks where each instance can belong to multiple categories simultaneously. This metric evaluates the proportion of actual positive labels correctly identified by the model and is particularly important in situations where identifying all relevant positive instances is critical. MR is calculated as: $\text{MR} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$

6 Technique overview with performance analysis

Sections 3 and 4 present models designed for mono-task and multi-task VRD-CU. This section examines the core techniques illustrated in Fig. 4—**Feature Representation**, **Cross-Modality Fusion**, **Model Architecture**, and **Pretraining**—emphasizing their functional

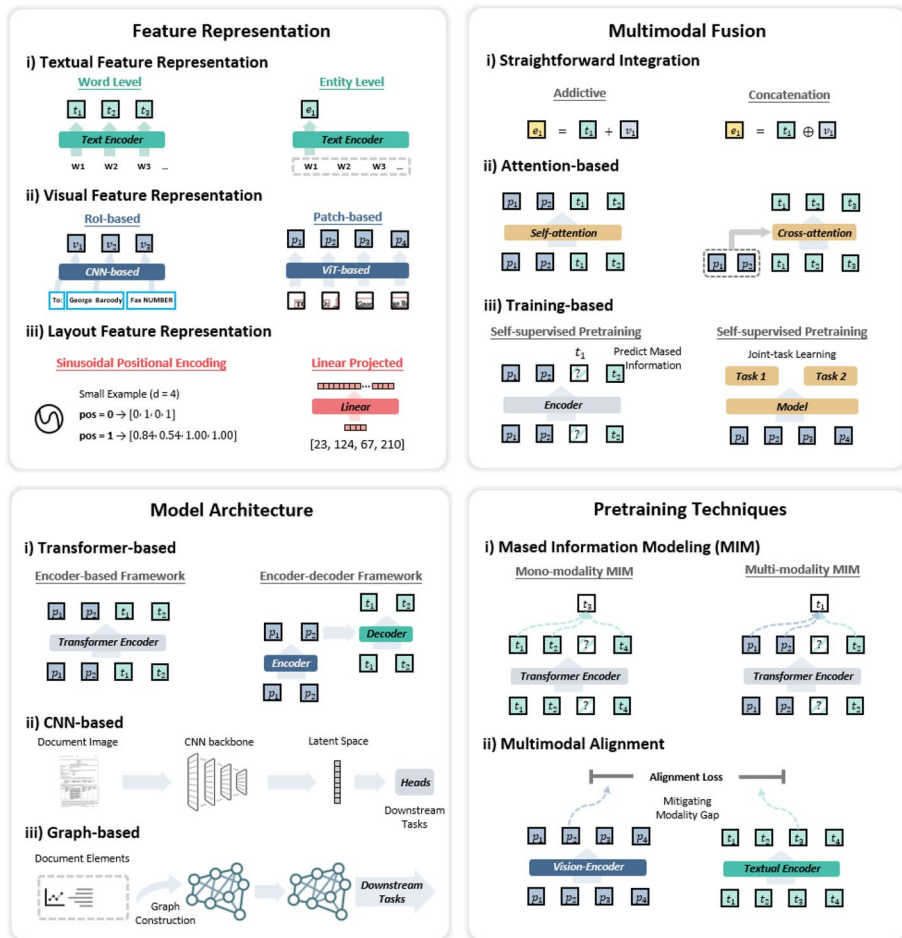


Fig. 4 Overview of VRD-CU techniques. It illustrates key aspects of multimodal feature representation and fusion, model architectures, and pretraining strategies for encoding and integrating textual, visual, and layout information in document understanding

roles, strengths, and limitations across diverse contexts. It further discusses recent progress in integrating **LLM and MLLM into VRD-CU** pipelines, highlighting emerging trends and paradigm shifts. Finally, to provide a comprehensive understanding of model performance, we summarize the quantitative results of representative approaches four broadly used VRD-CU benchmarks, offering a comparative view of their empirical effectiveness.

6.1 Feature representation

6.1.1 Textual representation

Text in VRDs provides essential semantic context, crucial for understanding content and conducting various downstream tasks. Depending on the information granularity required

by the framework and application scenarios, textual representation methods can generally be categorized at the word or entity level, as illustrated by Fig. 5.

Word-level Representations In VRDs, text sequences extracted by off-the-shelf OCR tools or PDF parsers (e.g., PDFMiner) can be encoded using word embedding methods such as Word2Vec (Mikolov et al. 2013) and Glove (Pennington et al. 2014). For more comprehensive textual embeddings, various BERT-style bi-directional pretrained transformer models like BERT (Devlin 2018) and RoBERTa (Liu et al. 2019) are employed to generate context-aware word representations. As the visually rich and structurally complex nature of VRDs, layout-aware and visual-integrated fine-grained models have been developed, such as LayoutLM families (Xu et al. 2020a, b; Huang et al. 2022; Xu et al. 2021), LiLT (Wang et al. 2022). These models generate word representations that integrate multimodal information, combining text, visual cues, and layout structure to achieve SoTA performance on several downstream tasks.

Entity-level Representations To acquire a dense representation of a text sequence for performing entity-level VRD-CU tasks, various approaches are adopted. These include averaging word embeddings specific to an entity or leveraging the [CLS] token to encapsulate the entire sequence, including averaging the word embeddings belonging to an entity or using [CLS] token to represent an entire sequence. SentenceBERT (Reimers and Gurevych 2019) is also often adopted to encode text sequences within document entities. However, a standardized approach for acquiring textual representations of entities is yet to be established, necessitating preliminary testing and validation.

6.1.2 Visual representation

Visual information provides layout, structural insights, and rich contextual clues, making it easier for humans to interpret and prioritize content and resulting in a more comprehensive reading experience. Based on the methods used for encoding visual information, we categorize them into two main types: CNN-based and Vision Transformer-based approaches, as represented by Fig. 6.

CNN-based Vision Encoding Methods involve first acquiring Region of Interest (RoI) bounding boxes and then applying RoI-pooling and RoI-Align on pretrained CNN backbones (e.g., Faster-RCNN or Mask-RCNN) to extract the region features. Many frameworks (Xu et al. 2020a; Li et al. 2021a; Luo et al. 2022; Ding et al. 2024; Zhai et al. 2023; Ding et al. 2023; Gu et al. 2021) utilize word or entity-level RoIs to effectively extract visual features of target regions and leverage the implicit knowledge embedded in pretrained backbones. However, acquiring the bounding boxes of words or entities incurs additional costs.

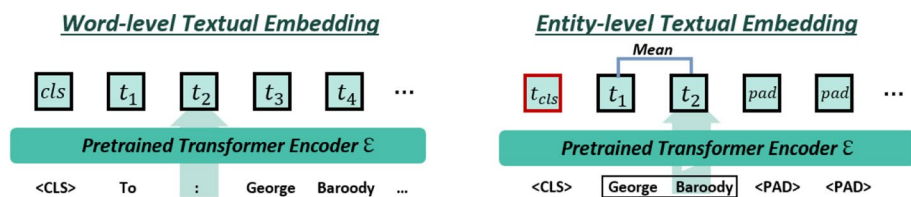


Fig. 5 Fine-grained versus coarse-grained textual embeddings in VRD-CU. Left: Word-level embedding uses $\mathcal{E}(\text{text}) = \{t_1, t_2, \dots, t_n\}$. Right: Entity-level embedding aggregates token vectors via mean pooling: $e = \frac{1}{k} \sum_{i=1}^k t_i$, where $\{t_1, \dots, t_k\}$ are tokens within an entity

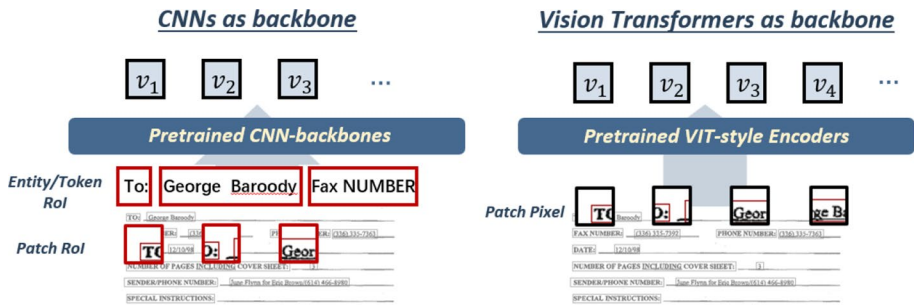


Fig. 6 Comparison of visual backbone architectures for document understanding. Left: CNN-based backbones extract features at the entity/token or patch level using region-of-interest (RoI) pooling. Right: Vision Transformer (ViT)-style encoders operate directly on fixed-size image patches, enabling patch-level feature extraction across the entire document. Both approaches utilize pretrained models to obtain visual embeddings $\{v_1, v_2, \dots\}$ for downstream tasks

Therefore, several frameworks (Xu et al. 2020b, 2021; Bai et al. 2022; Appalaraju et al. 2021) directly use image patch bounding RoIs to extract visual features and learn contextually with other modalities.

Transformer-based Vision Encoding After acquiring the visual features, they are typically fed into a transformer framework to fuse multimodal information, which can create significant computational bottlenecks. Additionally, acquiring high-quality RoIs of words or entities requires supervised training. To address these challenges, **LayoutLMv3**, inspired by ViT (Dosovitskiy et al. 2020), introduces a transformer-only framework. This approach applies a linear layer to project flattened patch pixels, which are then fed into a multimodal transformer to contextually learn with other modalities. This method reduces the number of parameters and simplifies the preprocessing steps, making it more efficient and adopted by many recent frameworks (Tang et al. 2022; Appalaraju et al. 2024; Mao et al. 2024). However, this encoding method cannot take advantage of implicit knowledge in pretrained frameworks and typically requires extensive pretrained.

6.1.3 Layout representation

Layout information is crucial for understanding document elements' spatial arrangement, including words and entities. Enhance document representation by clarifying the spatial relationships between these elements, thereby aiding in the comprehension of the overall document structure. The coordinates of the bounding box (bbox) of the document elements serve as initial layout information. This layout information can then be encoded using methods such as positional encoding, linear projection, and spatial-aware attention bias.

2D positional encoding, first introduced by **LayoutLM** (Xu et al. 2020a), and widely adopted by many VRD-CU models, allows the model to be aware of the relative spatial positions within a document. In this approach, document elements are normalized and discretized into integer ranges, and two separate embedding layers are used to encode the x and y coordinates, respectively. Despite its widespread use in models like (Xu et al. 2020a, b, 2021; Huang et al. 2022; Tu et al. 2023; Wang et al. 2022; Bai et al. 2022; Liao et al. 2023) this method encodes x and y coordinates individually, making it challenging to represent continuous 2D space and capture special correlations between document elements. Some

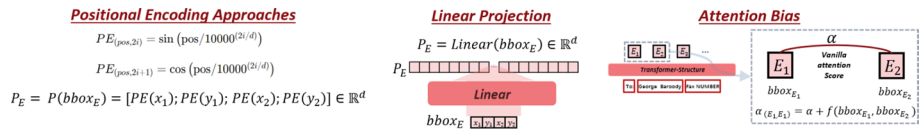


Fig. 7 Overview of layout encoding methods. Left: Sinusoidal positional encoding applied to bounding box coordinates $PE(\text{bbox}_E) = [PE(x_1); PE(y_1); PE(x_2); PE(y_2)] \in \mathbb{R}^d$. Middle: Linear projection of layout features $P_E = \text{Linear}(\text{bbox}_E) \in \mathbb{R}^d$. Right: Attention bias, where layout-aware attention is modified via $\alpha(E_1, E_2) = \alpha + f(\text{bbox}_{E_1}, \text{bbox}_{E_2})$

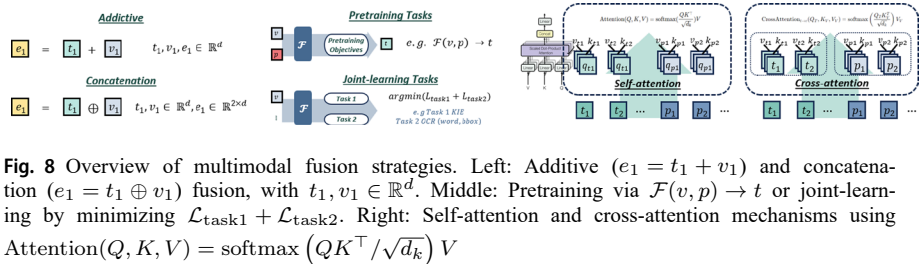


Fig. 8 Overview of multimodal fusion strategies. Left: Additive ($e_1 = t_1 + v_1$) and concatenation ($e_1 = t_1 \oplus v_1$) fusion, with $t_1, v_1 \in \mathbb{R}^d$. Middle: Pretraining via $\mathcal{F}(v, p) \rightarrow t$ or joint-learning by minimizing $\mathcal{L}_{\text{task1}} + \mathcal{L}_{\text{task2}}$. Right: Self-attention and cross-attention mechanisms using $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$

models (Ding et al. 2023; Wang et al. 2020) that follow the approach of **LXMERT** (Tan and Bansal 2019) utilize linear projection to update the x and y coordinates of the normalized bounding box coordinates simultaneously. To address the limitations of absolute positional encoding and incorporate relative positional correlations, other models introduce spatial-aware attention mechanisms (Hong et al. 2021; Appalaraju et al. 2021; Wang et al. 2022). These mechanisms enable vanilla self-attention to learn spatial dependencies effectively (Fig. 7).

6.2 Multi-modality fusion

After acquiring multimodal representations, it is important to explore effective fusion methods to integrate textual, visual, and layout information. This integration improves document understanding and boosts performance on downstream tasks. The straightforward integration methods, as shown in Fig. 8, include the additive and concatenation of the feature vectors. For example, additive integration sums layout information with corresponding textual or visual token representations (Xu et al. 2020a, b; Huang et al. 2022; Li et al. 2021a), while concatenation merges visual and textual features of document entities (e.g. tables) Ding et al. (2024, 2023). However, these methods require one-to-one correlations and alternative approaches are needed when such correlations are not available. Consequently, self-attention and cross-attention mechanisms are widely adopted to enhance each modality by learning inter- and intra-modality contexts. These mechanisms are commonly used in frameworks (Xu et al. 2020b; Huang et al. 2022; Tu et al. 2023; Bai et al. 2022; Xu et al. 2021) that integrate patch-level visual embeddings with textual features for contextual learning. Novel self-attention (Appalaraju et al. 2021; Wang et al. 2022) and cross-attention (Gu et al. 2021; Li et al. 2021a; Zhai et al. 2023) methods have been proposed to fuse multimodal information more effectively. Apart from model-based fusion approaches, self-supervised and joint learning tasks are also effective for integrating multi-aspect features. Self-supervised

vised pretraining tasks such as Masked Visual-Language Modeling (Wang et al. 2022; Xu et al. 2020a, b; Li et al. 2021; Huang et al. 2022; Zhai et al. 2023), Text-Image Alignment (Appalaraju et al. 2021), Text-Layout Pairing (Wang et al. 2022), and Text-Image Matching (Xu et al. 2020b, 2021) can significantly enhance multimodal information fusion. These methods require large-scale pretraining to learn cross-modality semantic correlations. Joint learning methods, often used in OCR-free frameworks (Wang et al. 2021; Kim et al. 2022; Davis et al. 2022; Yu et al. 2022), design auxiliary text detection or recognition tasks to fuse textual and visual information. This approach reduces pre-processing during inference and addresses mis-ordering sequence issues. However, these methods generally underperform compared to OCR-dependent models and involve additional training costs.

6.3 Model architecture

6.3.1 Transformer in VRD-CU

Referring to the models introduced in Sects. 3 and 4, transformers have become extensively utilized in VRD-CU tasks, attaining state-of-the-art performance due to several key advantages. Firstly, the attention mechanism effectively captures long-range dependencies within the multimodal information, including text, vision, and layout. Furthermore, the inherent scalability of transformers improves self-supervised learning on large-scale datasets (e.g. IIT-CDIP Lewis et al. (2006)), allowing them to handle diverse types and formats of documents, capturing more intricate and comprehensive features of the documents. Based on the transformer architecture used, VRD-CU models can be divided into two categories: encoder-only and encoder-decoder-based models. The first encoder-only model, LayoutLM (Xu et al. 2020a), was inspired by BERT (Devlin 2018) and uses various pretraining tasks to allow the bidirectional transformer encoder to capture more textual and layout information. Following LayoutLM, more pretrained VRD-CU models with encoder-only, layout-aware (Wang et al. 2022; Hong et al. 2021; Chen et al. 2022; Wang et al. 2022; Tu et al. 2023; Li et al. 2021) or visual integrated (Xu et al. 2020b, 2021; Appalaraju et al. 2021; Huang et al. 2022) have been proposed. These models are pretrained on various tasks to enhance their understanding of document structures. Encoder-only models demonstrate remarkable performance in sequence tagging and document classification tasks. However, they face challenges related to heavy annotation requirements and low readability, and they struggle with generative tasks such as abstractive question answering. Additionally, OCR errors can complicate the extraction of accurate information from input text. Furthermore, the fixed maximum input length of encoder-only frameworks limits their ability to handle long document inputs effectively. To overcome the limitations of encoder-only frameworks in generative QA and KIE, several encoder-decoder models (Powalski et al. 2021; Tang et al. 2022; Appalaraju et al. 2024; Mao et al. 2024) have been developed, but they still depend on costly OCR tools which can introduce errors affecting performance. OCR-free frameworks (Kim et al. 2022; Davis et al. 2022; Cao et al. 2023; Yu et al. 2022) address this issue by using vision encoders and text decoders for end-to-end processing. For long documents, T5-based encoder-decoder models (Raffel et al. 2020) have been proposed to effectively handle multi-page contexts.

6.3.2 CNNs in VRD-CU

In VRD-CU, CNNs are employed as the core framework for extracting feature maps from document images (Yang et al. 2017; Palm et al. 2019) and character grids (Katti et al. 2018), benefiting from their strong local feature extraction capabilities. Joint-learning frameworks (Wang et al. 2021) use CNNs as a backbone to integrate OCR and KIE tasks, combining visual and textual information through auxiliary tasks. Some OCR-free pretrained frameworks (Davis et al. 2022; Yu et al. 2022) also utilize CNNs as vision encoders to extract visual feature maps. However, CNNs struggle to capture long-range dependencies due to their localized receptive fields. To address this, StrucTexTv2 (Yu et al. 2022) combines CNN-extracted feature maps with a transformer to capture global contextual information. Additionally, CNNs are commonly used to extract visual features from regions of interest (RoI) with RoI Align (Xu et al. 2020a, b, 2021; Huang et al. 2022; Tu et al. 2023; Wang et al. 2022; Bai et al. 2022; Liao et al. 2023). Although CNNs can accurately capture region-specific visual cues and leverage pretrained knowledge from general domains, they require extra processing to obtain RoI bounding boxes, unlike vision transformers (Huang et al. 2022; Tang et al. 2022; Appalaraju et al. 2024; Mao et al. 2024), which operate directly on patch pixel values.

6.3.3 Graphs in VRD-CU

VRDs feature complex spatial and logical structures. The spatial structure shows the layout and positional relationships, such as a *Title* above a *Paragraph* and a *Caption* near a *Figure* or *Table*. The logical structure denotes semantic and hierarchical connections, like a *Title* being the parent of a *Paragraph* and a *Caption* describing a related *Table* or *Figure*. Graph-based frameworks explicitly encode these relationships using node and edge representations; therefore, GNNs are widely used in VRD-CU models (Lee et al. 2022; Shi et al. 2023; Luo et al. 2022; Zhang et al. 2021) to encode the spatial and logical representations. Although GNNs effectively capture domain-specific knowledge, they struggle with scalability and general domain knowledge pretraining. To address this, some frameworks (Li et al. 2023; Zhang et al. 2022) use attention masks or biases to mimic relationships between document elements, blending attention mechanisms with explicit relational modeling.

6.4 Pretraining mechanisms

By performing various pretraining tasks, a model can enhance its generalization ability through extensive datasets and prior training. Inspired by advances in pretraining language (Devlin 2018; Brown et al. 2020) and vision models (He et al. 2016; Dosovitskiy et al. 2020), numerous pretraining tasks for VRD-CU have been developed that are typically trained in large-scale document collections. This section will summarize the pretraining techniques and datasets commonly used for VRD-CU pretraining.

6.4.1 Pretraining tasks

Based on the purpose and pretraining targets, the pretraining methods can be categorized into Masked Information Modeling, Cross-modality Alignment, Mono-modality Augmentation, and Contrastive Learning.

Masked Information Modeling (MIM) is first introduced by Masked Language Modeling in BERT (Devlin 2018), which randomly masks 15% wordpiece tokens and requires the model to predict the masked tokens. Some models use multi-source inputs, such as XDoc (Chen et al. 2022) and MarkupLM (Li et al. 2022), which directly adopts *Masked Language Modeling* as a pretraining task on plain text or markdown text subsets. Some methods optimize MLM by changing masking wordpieces to whole words (*Whole Word Masking* by LayoutMask (Tu et al. 2023)) or all tokens belong to one randomly generated text block, named *Area-Masked Language Modeling* (Hong et al. 2021). Similar strategies can also be applied to mask entity-level textual representation, e.g. *Masked Sentence Modeling* (Gu et al. 2021). Those language-targeted masked information modeling to improve the language understanding ability of VRD-CU models. Except for language-focused masking strategies, vision (Gu et al. 2021; Yu et al. 2022) and layout-focused (Tu et al. 2023; Wang et al. 2022) strategies are also adopted to learn the specific modality aware general knowledge.

Moreover, *MIM* is an effective method to boost cross-modality understanding. For example, LayoutLM (Xu et al. 2020a) introduces a *Masked Visual-Language Modeling* which allows using kept visual/layout information and contextual text content to predict the masked word-pieces, adopted by many VRD-CU pretrained models (Wang et al. 2022; Xu et al. 2020b; Li et al. 2021; Huang et al. 2022; Zhai et al. 2023; Appalaraju et al. 2021). Similarly, some visual token masked models leverage multimodal information to reconstruct the view tokens, e.g. *Learn to Reconstruct* (Appalaraju et al. 2021), *Masked Image Modeling* (Huang et al. 2022). Additionally, some models mask multimodal features simultaneously to conduct a cross-modality masking (Li et al. 2021a; Liao et al. 2023). Masking Information Modeling may have limitations on bias in masking strategies, thus some frameworks (Tu et al. 2023) may try different masking ratios or strategies to improve the training effectiveness. Additionally, other common concerns about *MIM* also include training efficiency and lack of structural information. Thus, other pretraining methods are introduced to mitigate the limitations.

Cross-modality Alignment Although some *Masked Information Modeling* methods could effectively boost the intra/inter-modality understanding, implicit contextual learning is limited to capturing explicit alignment between different modalities. Thus, few cross-modality aligning methods are introduced to enhance the modality interaction. To enhance the text-image interactive learning, Text-Image Alignment is adopted (Xu et al. 2020b, 2021) which covers the image region of token lines to predict whether the image region of the target token line is covered or not. LayoutLMv3 (Huang et al. 2022) expands from covering the image region only to image/text tokens to further enhance interactive learning. Vision-language alignment (Gu et al. 2021) and Text-Image Matching (Xu et al. 2020b, 2021) target to predict whether image-text features belong to the same region or not. DocFormer (Appalaraju et al. 2021) tends to predict the text content of the paired images.

Other Pretraining Techniques Some pretraining tasks are introduced to further enhance the understanding of specific modalities. To enhance layout information understanding,

StructuralLM (Li et al. 2021) and WOKONG-READER (Bai et al. 2022) introduce *Cell Position Classification* and *Textline Grid Matching* to predict which located grids are of each cell or textline. Fast-StrucText (Zhai et al. 2023) introduces a graph-based token relation method to predict a spatial correlation between token pairs. MarkupLM (Li et al. 2022) leverage the benefits from markup files to predict the logical parent–child relation between nodes by introducing a *Node Relation Prediction*. Contrastive learning-based strategies are adopted to conduct single (Gu et al. 2021) or cross-modality contrastive (*Textline-Region Contrastive Learning*) (Bai et al. 2022) learning.

6.4.2 Pretraining datasets

To perform the aforementioned tasks, large-scale document collections are essential for conducting self-supervised learning. Different pretraining datasets are adopted by various models. The most widely used pretraining dataset is the IIT-CDIP Test Collection 1.0 (Lewis et al. 2006), which contains more than 6 million documents with over 11 million scanned document images. Since it contains a cross-domain and large number of unannotated documents, it is used by the majority of models (Li et al. 2022; Hong et al. 2021; Kim et al. 2022; Xu et al. 2020a, b; Huang et al. 2022; Xu et al. 2021; Bai et al. 2022; Appalaraju et al. 2021; Liao et al. 2023; Wang et al. 2022; Gu et al. 2021; Zhai et al. 2023; Li et al. 2021; Luo et al. 2023; Chen et al. 2022; Appalaraju et al. 2024). As the original IIT-CDIP dataset provides the text content without layout information, off-the-shelf OCR tools are normally used to acquire the bounding box information of each document. Some models (Li et al. 2021b, a; Wang et al. 2022) use relatively smaller pretraining datasets like RVL-CDIP (Harley et al. 2015), which contains 400,000 evenly distributed documents in 16 types, to reduce the cost of pretraining. The multi-source model XDoc (Chen et al. 2022) also leverages many plain text corpora for pretraining, such as BookCORPUS, CC-NEWS, OPENWEBTEXT, STORIES and HTML-sourced CommonCrawl datasets. To address multilingual scenarios, both LiLT (Wang et al. 2022) and LayoutXLM (Xu et al. 2021) follow the principles and policies of Common Crawl to gather large amounts of multilingual digitally-born PDF documents.

6.5 LLM and MLLM into VRD-CU

In addition to frameworks specifically tailored for VRD-CU tasks, a growing number of general-purpose MLLMs have been proposed. These models aim to handle a wide range of modalities and domains—including plain text, natural scenes, documents, charts, and tables—and have demonstrated strong performance on various VRD-CU benchmarks. Many of these models incorporate specialized architectural mechanisms, pretraining or instruction tuning to support broad vision-language understanding across tasks. These general-purpose models can be broadly classified into two categories based on their reliance on text extraction: OCR-dependent and OCR-free LLMs/MLLMs.

6.5.1 OCR-dependent LLMs/MLLMs

OCR-dependent LLMs/MLLMs rely on off-the-shelf OCR tools to extract text and layout information, typically in the form of tokens and bounding boxes, from scanned or digital documents. These extracted elements are either directly embedded into LLM prompts (e.g.,

ICL-D3IE He et al. (2023), DocLLM Wang et al. (2024)) or fused with visual features using multimodal encoders like ViT (Dosovitskiy et al. 2020) or LayoutLMv3 (Huang et al. 2022) (e.g., LayoutLLM Luo et al. (2024), DoCo Li et al. (2024), GPE Zhu et al. (2025)). To incorporate layout cues, models commonly use 2D positional encoding, bounding box-aware prompts, or auxiliary encoders, with some frameworks verbalizing spatial coordinates or using layout-specific tokens to enrich input representations. While this modular design reduces the need for costly end-to-end pretraining on text recognition, it introduces vulnerabilities: OCR errors, especially in noisy or handwritten documents, can cascade through the pipeline, and unordered extracted text often requires additional processing like XY-cut reordering. Furthermore, layout representations are often implicit or rule-based (Lamott et al. 2024; He et al. 2023), which may fail to capture fine-grained structural dependencies. The use of low-resolution images to reduce computation can also limit the expressiveness of visual features, leading to degraded multimodal fusion and overall performance, especially on layout-sensitive tasks.

6.5.2 OCR-free LLMs/MLLMs

OCR-free LLMs/MLLMs have emerged as a promising direction when applying in end-to-end VRD-CU, eliminating the dependency on explicit text extraction by operating directly on document images. These models typically employ one or more vision encoders to extract visual features, which are then fused with user queries and decoded by LLMs to generate task-specific outputs such as, Donut, mPLUG-DocOwl, and UReader (Kim et al. 2022; Ye et al. 2023, 2023). Achieving fine-grained textual understanding in OCR-free settings requires high-resolution input, which results in long visual sequences and necessitates the use of visual compression modules to maintain computational tractability (Liu et al. 2024; Hu et al. 2024). Additionally, effective integration of textual semantics and document layout often hinges on large-scale pretraining. This is typically achieved through auxiliary tasks such as text recognition (Liu et al. 2024) and image captioning (Feng et al. 2024), which facilitate multimodal alignment. However, the paradigm is heavily reliant on extensive dataset construction and significant computational resources, presenting considerable challenges for scalability and real-world deployment.

6.6 Performance analysis

Table 3 provides a structured comparison of key VRD-CU models, enabling us to identify performance trends, limitations, and trade-offs across different architectural designs, task categories, and dataset domains.

Text-only versus Multimodal Models Early baselines such as BERT (Devlin 2018) and RoBERTa (Liu et al. 2019), while foundational for natural language understanding, perform worse on visually-rich document tasks due to their inability to model spatial and visual structures. On tasks like KIE (e.g., FUNSD) and document QA (e.g., DocVQA), these text-only models underperform by about 15–30 percentage points compared to multimodal alternatives. This performance gap underscores the importance of incorporating layout (L) and visual (V) modalities alongside text (T). LayoutLMv3 (Huang et al. 2022) and GeoLayoutLM (Luo et al. 2023) exemplify this shift, offering gains of over 10 points on datasets such as SROIE and XFUND through layout-aware and multilingual pretraining.

Task-specific versus Multi-task Models Mono-task models like SPADE (Hwang et al. 2021) and FormNetv2 (Lee et al. 2023) are tailored for KIE, often achieving high performance on structured forms (e.g., 98.2% on SROIE). However, these models may show reduced performance on QA or classification tasks, indicating limited generalization capacity. In contrast, multi-task frameworks such as Wukong (Bai et al. 2022) and UDoc (Gu et al. 2021) demonstrate broader versatility, achieving consistently high scores across both KIE and QA tasks (e.g., UDoc reports 98.94% on CROD and 85.3% on DocVQA). This may suggest that task-agnostic pretraining with cross-modal alignment confers better robustness and generalization, particularly when applied to real-world, multi-format documents.

OCR-based versus OCR-free Models OCR-dependent models remain dominant in terms of accuracy on text-intensive tasks. For instance, LayoutLMv3 achieves over 80% on DocVQA and near-perfect accuracy on SROIE due to reliable text extraction. Meanwhile, OCR-free models like Donut (Kim et al. 2022) and Dessurt (Davis et al. 2022) show promising results in terms of end-to-end pipeline simplification and robustness to OCR noise, but they generally fall short by 5–15 percentage points on benchmarks such as FUNSD and XFUND. This performance drop is particularly notable for multilingual or handwritten datasets where precise text recognition remains a bottleneck. Therefore, OCR-free models may offer an elegant alternative but require further advances in visual text understanding to close the performance gap.

Model Capacity and Efficiency Trade-offs Models like Wukong-large and LayoutMask-large demonstrate that increasing model capacity enhances performance, especially on high-complexity tasks, with LayoutMask-large reaching 93.2% on FUNSD. However, such improvements come at the cost of computational demands, memory usage, and slower inference, which may hinder practical deployment in resource-constrained settings. Lightweight models or those leveraging parameter-efficient tuning (e.g., adapters, LoRA) are emerging as viable solutions, though they currently underperform on complex multi-modal reasoning tasks.

In summary, performance trends across VRD-CU models reveal that (1) multimodal integration consistently outperforms text-only approaches, (2) multi-task learning may improve cross-domain generalization, (3) OCR-free methods offer architectural simplification at the cost of task accuracy, and (4) higher model capacity often improves performance but exacerbates scalability and deployment challenges. These observations not only provide critical insights into the trade-offs inherent in current designs but also guide future directions in balancing performance, generalizability, and efficiency.

7 Critical discussion about future trend and practical challenges

7.1 Challenges in multimodal information representation

The inherently multimodal nature of VRD-CU tasks is well established (Ding et al. 2025), with recent frameworks increasingly integrating multimodal information through varied strategies. In the text modality, a primary challenge lies in accurately extracting content from scanned and handwritten documents. Both OCR-dependent (Huang et al. 2022; Appalaraju et al. 2024) and OCR-free methods (Kim et al. 2022; Davis et al. 2022) continue to struggle with this issue, which remains contingent on advances in text spotting and recognition.

Table 3 (continued)

Model attributes				Architectural details					Performance scores									
Model	Venues	Year	Var	Size	PT	Tasks	Modality	OCR	Frmwk.	Grained	V.	L.	Fusion	FUNSD	SROIE	CROD	DocVQA	
					Size													
DocTR	CVPR	2023	–	153 M	Y	KIE	11 M	T+L+V	Y	T-E/T-D	Fine	CNN	PE	Cross-Att	84.0	98.2	/	/
Layout-aware pretrained frameworks																		
LayoutLM	KDD	2020	base	160 M	Y	QA, KIE	11 M	T+L	Y	T-E	Fine	N/A	PE	Concat	78.66	94.67	94.72	69.79
																Xu et al. (2020b)	Xu et al. (2020b)	
			Large	343 M		LA, DC									78.95	95.24	94.93	72.59
																Xu et al. (2020b)	Xu et al. (2020b)	
StructuralLM	ACL	2021	–	355 M	Y	KIE, QA, DC	11 M	T+L	Y	T-E	Coarse	N/A	PE	Self-Att	85.14	/	/	83.94
XDoc	EMNLP	2022	–	146 M	Y	KIE, LA, DC	42 M+ *	T+L	Y	T-E	Fine	N/A	PE	Sum	89.4	/	/	72.7
BROS	AAAI	2022	Base	110 M	Y	IE, RP	10.6M	T+L	Y	T-E	Fine	N/A	PE	Self-Att.	83.05	96.28	95.73	/
																Wang et al. (2022)	Tu et al. (2023)	/
			large	340 M		IE, RP									84.52	96.62	97.4	Tu et al. (2023)
																Wang et al. (2022)		/
LiLT	ACL	2022	–	–	Y	KIE, LA, DC	11 M	T+L	Y	T-E	Fine	N/A	PE	Cross-Att	88.41	/	96.07	/
LayoutMask	ACL	2023	Base	182 M	Y	KIE, DC	10 M	T+L	Y	T-E	Fine	N/A	PE	Self-Att.	92.91	96.87	96.99	/

Table 3 (continued)

Model attributes				Architectural details					Performance scores									
Model	Venues	Year	Var	Size	PT	Tasks	PT	Modality	OCR	Frmwk.	Grained	V.	L.	Fusion	FUNSD	SROIE	CROD	DocVQA
Vision-integrated pretrained frameworks																		
Fine-grained frameworks																		
LayoutLMv2	ACL	2021	Base	200 M	Y	QA, KIE	Y	T+L+V	Y	T-E	Fine	CNN	PE	Concat	82.76	96.25	94.95	78.08
			Large	426 M	LA, DC													
DocFormer	ICCV	2021	Base	183 M	Y	KIE, LA, DC	Y	T+L+V	Y	T-E	Fine	CNN	PE	Self-Att.	83.34	/	96.33	/
			Large	536 M														
LayoutLMv3	MM	2022	Base	133 M	Y	QA, KIE	Y	T+L+V	Y	T-E	Fine	ViT	PE	Self-Att.	90.29	/	96.56	78.76
			Large	368 M	LA, DC													
Coarse-grained frameworks																		
SelfDoc	CVPR	2021	-	-	Y	KIE, LA, DC	Y	T+L+V	Y	T-E	Coarse	CNN	PE	Cross-Att	83.36	/	/	/
UDoc	NeurIPS	2021	-	272 M	Y	KIE, LA, DC	Y	T+L+V	Y	T-E	Coarse	CNN	Linear	Cross-Att	87.93	/	98.94	/
MGDoc	EMNLP	2022	-	203 M	Y	KIE, DC	Y	T+L+V	Y	T-E	Coarse	CNN	PE	Concat	89.44	/	97.11	/
Joint-grained frameworks																		
StructText	MM	2021	-	107 M	Y	KIE, RP	Y	T+L+V	Y	T-E	Joint	CNN	Linear	Self-Att	83.09	96.88	/	/
mmLayout	MM	2022	Base	236 M	Y	KIE, LA	Y	T+L+V	Y	Graph	Joint	CNN	PE	Concat	86.02	97.63	97.23	79.15

Table 3 (continued)

Model attributes				Architectural details					Performance scores									
Model	Venues	Year	Var	Size	PT	Tasks	PT Size	Modality	OCR	Frmwk.	Grained	V.	L.	Fusion	FUNSD	SROIE	CROD	DocVQA
Fast-Struc- Text	IJCAI	2023	–	116 M	Y	KIE, RP	11 M	T+L+V	Y	T-E	Joint	CNN	PE	Cross-Att	86.49	97.91	97.38	83.66
	Wukong	ACL	2023	Base	237 M	Y	KIE, LA, DC	11 M	T+L+V	Y	T-E	Joint	ViT	PE	Self-Att.	91.52	96.88	96.54
GeoLay- outLM	CVPR	2023	–	399 M	Y	KIE, RP	11 M	T+L+V	Y	T-E	Joint	ViT	PE	Cross-Att	93.62	98.15	97.27	/
	ICDAR	2021	Base	230 M	Y	QA, KIE	1.1M	T+L+V	Y	T-E/T-D	Fine	CNN	PE	Concat	/	97.65	95.11	83.92
Dessurt	ECCV	2022	–	–	Y	KIE, QA, DC	N/A	T+V	N	T-E/T-D	Fine	CNN	N/A	Cross-Att	/	98.1	96.33	87.05
	ECCV	2022	–	176 M	Y	KIE, QA, DC	13 M	T+V	N	T-E/T-D	Fine	Swin	N/A	Cross-Att	/	/	84.1	72.1
SeRun	ICCV	2023	–	136 M	Y	KIE, QA	13 M+	T+V	N	T-E/T-D	Fine	Swin	N/A	Cross-Att	/	85.8	84.9	77.9
UDOP	CVPR	2023	–	742 M	Y	KIE, QA, DC	11 M	T+L+V	Y	T-E/T-D	Fine	ViT	PE	Concat	91.62	/	97.58	84.7
ViTLP	NAACL	2024	–	253 M	Y	KIE, QA, DC	13 M	T+L+V	Y	T-E/T-D	Fine	ViT	PE	Cross-Att	87.61	/	95.59	65.9
LLM-based frameworks																		

Table 3 (continued)

Model attributes				Architectural details							Performance scores							
Model	Venues	Year	Var	Size	PT	Tasks	PT	Modality	OCR	Frmwk.	Grained	V.	L.	Fusion	FUNSD	SROIE	CROD	DocVQA
LapDoc	ICDAR	2024	–	–	N	KIE, QA	N/A	T+L	Y	T-D	Fine	N/A	Prompt	Prompt	/	79.9	/	79.8
DocLLM	ACL	2024	–	7B	Y	KIE, QA, DC	3.8M	T+L	Y	T-D	Fine	N/A	PE	Cross-Att	51.8	91.9	67.4	69.5
LayoutLLM	CVPR	2024	–	7B	Y	KIE, QA	6 M+	T+L+V	Y	T-E/T-D	Fine	Fine	PE	Self-Att	79.98	72.12	63.1	74.27
HRVDA	CVPR	2024	–	7.2B+	Y	KIE, QA	13 M+	T+L+V	Y	VT-E/T-D	Fine	Swin	PE	LoRA	/	91	89.3	72.1
DocLayLLM	CVPR	2025	–	8B	Y	KIE, QA	3.1M	T+L+V	Y	VT-E/T-D	Fine	ViT	Prompt	Self-Att	84.11	84.36	71.34	78.36
LayTo-kenLLM	CVPR	2025	–	8B	Y	KIE, QA	5.3M	T+L	Y	T-D	Fine	N/A	PE	LoRA	81.62	/	78.3	85.11
GPE	ICLR	2025	–	7B	Y	KIE, QA	0.5M	T+L	Y	T-D	Fine	N/A	PE	Sum	82.6	97.8	86.9	78.1

This table provides a comprehensive overview of various VRD-CU models, detailing their published venue, architectural variant (Var), and pretraining (PT) size. It outlines their supported Tasks, including Key Information Extraction (KIE), Question Answering (QA), Layout Analysis (LA), Document Classification (DC), and Relation Prediction (RP). The table also specifies their input Modality (Text (T), Layout (L), Visual (V)), Optical Character Recognition (OCR) tool requirement, and Framework (Frmwk.) types (e.g., Transformer Encoder (T-E), Self-attention (Self-atten.), Transformer Decoder (T-D)). Further architectural details include granularity levels (Grained), visual encoder (V), layout representation (L.) (e.g., Linear projected (Linear), Position Encoding (PE)), and fusion mechanisms (e.g., Concatenation (Concat), Self-attention (Self-Att), Cross-Attention (Cross-Att.)). Performance scores are reported on the FUNSD, SROIE, CROD, and DocVQA datasets, using F1-score for FUNSD, SROIE, and CROD, and ANLS for DocVQA. For papers where the original work did not report the performance, we cite the reproducing paper in the upper-right corner of the corresponding performance value

Another key limitation is the application of these frameworks to low-resource languages, where the scarcity of annotated datasets (Xu et al. 2021) and the reduced performance of LLMs significantly undermine effectiveness. On the visual side, most pretrained VRD-CU models rely on low-resolution images to limit the number of visual tokens, often at the cost of fine-grained details—particularly problematic for OCR-free systems. Although high-resolution-compatible methods (e.g., adaptive cropping (Ye et al. 2023), visual compressors (Hu et al. 2024), and resampler (Wang et al. 2024)) have emerged, they face persistent trade-offs between token efficiency and content preservation, especially when handling lengthy documents. Finally, layout encoding has evolved from simple 2D positional encoding to RoPE-based methods (Zhu et al. 2025) compatible with LLMs. Recent approaches increasingly embed layout features during pretraining and instruction tuning rather than relying solely on prompts or external inputs. Nonetheless, scaling layout encoding to long documents remains difficult, particularly in preserving spatial coherence across pages.

7.2 Challenges in multimodal information fusion

Beyond challenges in feature representation, the effective fusion of encoded multimodal features remains an open research problem. Direct fusion methods, such as simple summation or concatenation, lack the capacity to capture complex inter-modal dependencies and often underperform in scenarios requiring fine-grained alignment. Additionally, as model sizes grow and input sequences expand—especially due to the incorporation of fine-grained visual and layout features—self-attention and cross-attention mechanisms become computationally expensive, posing scalability challenges for large-scale pretraining and instruction tuning in LLM-based frameworks. To mitigate this, lightweight modules such as adapters and LoRA have been adopted to reduce training costs (Feng et al. 2024; Yu et al. 2024a, b). However, LoRA-style methods often exhibit limited flexibility and generalization, requiring task-specific supervised fine-tuning and performing inconsistently across diverse document types. Additionally, while several well-defined pretraining and instruction-tuning tasks have been proposed to enhance multimodal fusion especially for OCR-free frameworks (Wei et al. 2024; Wang et al. 2025; Zhang et al. 2024), even LLM-based models trained on large-scale datasets for text spotting and recognition still underperform compared to OCR-dependent or small-scale pretrained VRD-CU models.

7.3 Challenges with multi-page and long documents

In real-world scenarios, VRDs often span multiple pages—such as academic papers, financial statements, or industrial reports—yet most existing frameworks are designed for single-page inputs and are constrained by token limits (e.g., 200 visual and 512 textual tokens in LayoutLM-series (Xu et al. 2020a, b; Huang et al. 2022; Xu et al. 2021)), making it difficult to process text-dense page or multi-page documents. Existing multi-page approaches (e.g., Tito et al. (2023); Blau et al. (2024)) typically rely on retrieval mechanisms to select relevant pages, which are then processed independently by MLLM-based VRD-CU systems; however, such pipelines often fail to capture semantic and logical dependencies across pages, leading to incomplete contextual understanding. While Hu et al. (2024) introduce multi-page scale pretraining to enhance cross-page comprehension, their focus remains primarily on text recognition, with limited modeling of semantic and logical relationships.

Additionally, managing long input sequences—especially with high-resolution images—poses challenges for effective multimodal retrieval, compression, and fusion, as maintaining spatial and logical coherence across long text-visual sequences remains difficult. Moreover, existing multi-page benchmarks (Ding et al. 2024; Van Landeghem et al. 2023; Tito et al. 2023) are mostly limited to extractive tasks with low inter-page complexity, leaving critical challenges like multi-hop and multimodal reasoning under-explored.

7.4 Challenges in data efficiency: synthetic, few/zero-shot learning

In real-world scenarios, acquiring high-quality, manually curated datasets for new document collections is time-consuming and labor-intensive, often requiring domain experts to annotate and ensure data quality (Gandhi et al. 2024; Ding et al. 2025). While a few studies (Chen et al. 2023; Wang and Shang 2022) have proposed few-shot learning frameworks, they remain limited—particularly in KIE tasks—with constrained sets of pre-defined keys and generally underperform compared to fully fine-tuned models. To address data scarcity and improve few-shot and zero-shot performance, recent LLM/MLLM-based approaches (Touvron et al. 2023; Hu et al. 2024) have turned to synthetically generating large-scale instruction-response pairs for instruction tuning. This strategy helps models better understand user queries in low-resource settings, yet noticeable performance gaps persist, as shown in Table 3. A recent study (Ding et al. 2024) further investigates tuning models with synthetic datasets to bridge domain gaps between general-domain pretraining and downstream document tasks. However, synthetic data often lacks the quality, complexity, and diversity of manually curated datasets (Gandhi et al. 2024), limiting its effectiveness for robust few-shot and zero-shot generalization. As such, future work should explore active learning, human-in-the-loop pipelines and reinforcement learning to improve the authenticity, task alignment, and transferability of synthetic training data (Li et al. 2024).

7.5 Challenges in domain-specific applications

While most benchmark-driven VRD-CU research has focused on general-purpose or synthetic datasets, real-world deployment in domain-specific settings introduces a host of practical challenges that remain under-explored. In fields such as law, finance, healthcare, and robotic process automation (RPA), documents often exhibit unique structural conventions, sensitive content, and operational constraints that demand specialized adaptation. A summary of these domain-specific challenges and their implications is presented in Table 4.

One critical issue is the presence of OCR noise in low-quality scans, handwritten forms, or degraded faxes, which are particularly common in archival legal documents or hospital records. Although recent OCR-free models demonstrate robustness under certain conditions, they still struggle with noisy inputs, particularly in multilingual or script-heavy environments. For example, forms from multilingual regions, such as the EU or India, often contain mixed scripts (e.g., Hindi-English or French-German), making tokenization, layout modeling, and language alignment significantly more complex. Despite promising efforts such as LiLT (Wang et al. 2022) and XFUND (Xu et al. 2021), limitations in cross-lingual pretraining and a lack of high-quality multilingual annotations hinder progress.

Latency and computational cost are additional concerns for latency-sensitive applications, such as RPA workflows or mobile-based document scanning tools. Transformer-based

Table 4 Summary of domain-specific challenges in VRD-CU applications

Aspect	Description	Affected domains	Remarks
OCR Noise	Noisy scans, handwriting, and low-resolution inputs affect text quality	Legal, Healthcare	OCR-free models still struggle with poor-quality or multilingual handwritten inputs
Multilinguality	Mixed scripts and cross-lingual layout inconsistencies	Government, Finance, Education	Limited multilingual pretraining; lack of annotated cross-lingual datasets
Latency and Efficiency	Long documents and high-resolution inputs increase inference cost	RPA, Mobile Apps, Enterprise Systems	Transformer-based models have high memory/latency demands; few studies address real-time use
Privacy and Data Sensitivity	Sensitive information restricts data use and sharing	Healthcare, Finance	Requires privacy-preserving learning (e.g., federated learning) and explainability
Domain Adaptation	Models fail to generalize across domains with unique templates and semantics	Legal, Insurance, Logistics	Domain-adaptive pretraining and few-shot adaptation remain under-developed
Explainability	Lack of interpretable outputs for extracted fields or decisions	Legal, Healthcare, Finance	Needed for compliance, auditability, and user trust

VRD-CU models, particularly those requiring high-resolution images or long-sequence attention, pose challenges in real-time scenarios due to memory and processing overhead. Lightweight alternatives, such as LoRA or early-exit mechanisms, have not yet been comprehensively validated in production-grade settings where speed, interpretability, and robustness must be balanced. Moreover, in regulated domains such as finance and healthcare, data privacy and explainability are essential. The use of large-scale pretrained models raises concerns around data leakage, reproducibility, and the opaque nature of predictions. Current VRD-CU systems often lack mechanisms to provide transparent rationales for extracted content or document-level decisions, posing a barrier for adoption in high-stakes settings.

Overall, greater attention must be paid to domain-specific customization, multilingual document handling, and real-world constraints such as noise, latency, and privacy. Future directions include domain-adaptive pretraining, federated learning for privacy-preserving modeling, and the development of cross-lingual VRD-CU benchmarks that reflect the diversity and complexity of real-world documents. We also present Table 5, which illustrates how different classes of VRD-CU models are aligned with domain-specific applications such as finance, healthcare, multilingual administration, and archiving.

Table 5 Mapping of model classes to real-world VRD-CU applications

Model class	Representative models	Practical application domains
Layout-aware Transformers	LayoutLM, LayoutLMv3, DocFormer	Invoice analysis, form classification, information extraction in finance and healthcare
OCR-free Vision-Language Models	Donut, DES-SURT, SimulDoc	Digital archiving of historical documents, handwritten note recognition, multi-script forms
Multilingual and Cross-lingual Models	LiLT, LayoutXML, Xfund-adapted LayoutLMv3	Processing multilingual government forms, EU compliance documents, Indian tax filings
High-Resolution Document Models	mPLUG-Owl, UReader, Qwen-VL	Industrial reports, technical diagrams, patents with dense layout or large images
Instruction-tuned Multimodal LLMs	DocPedia, mPLUG-DocOwl, DocGLM-IT	Robotic Process Automation (RPA), document QA, interactive document agents

8 Conclusion

This paper comprehensively reviews deep learning-based models for visually rich document content understanding (VRD-CU), encompassing both mono-task frameworks designed for specific VRD-CU downstream tasks and multi-task frameworks that support multiple VRD-CU downstream tasks. Beyond introducing the novelties of each model, the limitations of these frameworks are summarized at the end of each section, offering a thorough trend analysis. Additionally, this paper summarizes existing VRD content understanding datasets, pointing out future trends and demands for VRD-CU. To provide a systematic review, we critically discuss various techniques, highlighting their strengths and limitations. Due to accessibility restrictions, proprietary industrial systems are not fully incorporated in this survey. Meanwhile, the continuous advancement of multimodal large language models is anticipated to substantially expand the analytical perspectives and application boundaries of the VRD-CU field. Nevertheless, we believe this work provides a comprehensive and timely overview of the development of VRD-CU, serving the interests of both academic research and industrial application.

Author Contributions Yihao Ding led the conceptualization and writing of the manuscript, conducted the literature review, and organized the taxonomy of VRDU frameworks and datasets. Soyeon Caren Han supervised the project, contributed to the critical revisions, and coordinated the collaboration between author teams. Jean Lee contributed to the related work section by surveying previous studies and assisted in manuscript checking and proofreading. Eduard Hovy provided high-level guidance on structuring the survey, critically reviewed the manuscript, and contributed to the framing of future research directions.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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